

# Sound Insulation of Double Walls – An Experimental Parametric Study

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## Summary

The sound insulation of double panels has been widely studied in the literature so far. In most previous studies, only one or two double wall parameters are dealt with at a time and the overall comprehension of the behaviour of the double panel can be lost. For example, studs are often ignored. The purpose of this study is to strengthen our all-inclusive understanding of double panels. The study was implemented by means of a carefully planned experimental parametric study. Altogether 54 sound insulation tests were done with double panels where the cavity properties were changed. The panel material was 2-mm thick steel to obtain a high critical frequency. The most important parameters for uncoupled structures were the amount of absorbent and the cavity thickness. The flow resistivity of the mineral wool did not play a significant role. For coupled structures, the stiffness (or type) of the stud, and the spacing of the screws between the panel and the stud were the most important parameters. Flexible studs were better than rigid studs at low and middle frequencies but the difference was smaller at high frequencies than expected. The stud spacing did not play a significant role except at low frequencies, when dense stud spacing was used. In this case, very strong low frequency resonance occurred. The resonance is probably caused by horizontal normal modes of the sub-panels, which are formed between vertical studs. This resonance and the effects of stud stiffness and screw spacing should be further investigated in the future.

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## 1. Introduction

The sound insulation behaviour of thin single panels is rather well-known. The airborne sound insulation of single panels depends on the surface mass, Young's modulus, dimensions, total loss factor and Poisson's ratio. For thick panels, the shear modulus is also needed to account for the transverse waves at high frequencies.

Multiple walls are used when lightweight but sound-insulating structures are required, e.g., in doors, windows and mobile walls. They are also used as a primary type of partition wall in re-fitting, repair, offshore structures, ships and high buildings. In some countries, lightweight walls are used as a standard partition wall between dwellings.

The sound insulation behaviour of double walls is much more complex than that of single panels. To predict the sound insulation, in the first place, the properties of the constituent single panels are needed. In addition, one needs to know the distance between the panels, the properties of the absorbent and the properties of the interpanel connections (studs, joists or purlins). Solid interpanel connections between the panel and the stud can act like a point or a line bridge depending on the distance between the

screws or other fasteners. The density of interpanel connections and their dynamic stiffness have to be known.

The absorbent can also be characterized by several parameters. When the mechanical coupling via the frame of the cavity absorbent is negligible (unbonded absorbent), the amount of the absorbent, porosity, density and flow resistivity are expected to be important. For sandwich panels, where the absorbent is bonded to the panels, Young's modulus, Poisson's ratio, loss factor, and structure factor can also be important.

Several studies can be found in the literature where one or several of the above-mentioned parameters have been studied. They are also reviewed below. However, it is difficult to form an overall comprehension of the effects of these parameters, or their combined effects, on sound insulation. In addition, some textbooks give partially oversimplified views, which are inconsistent with practice. There is a lack of a study where the most important parameters are studied simultaneously without changing panel properties, and where all material parameters are properly documented.

The purpose of this study was to improve our understanding of the sound insulation of double panels. This is implemented by presenting an experimental parametric study of double panels where double wall parameters are systematically varied, and their effect on sound insulation is graphically presented. The second purpose was

to produce reliable data for the validation of double panel prediction models. A preliminary study by Hongisto revealed that most of the existing models have been validated with wall structures, which have been inadequately documented [1, 2]. Both verification and validation of the existing models are difficult. The next step is to make an extensive comparison between the prediction models utilizing the experimental data of this study.

## 2. Literature review

In the following, a review of the literature is presented concerning the influence of different physical parameters on the SRI (sound reduction index) of double panels. The review starts from uncoupled structures where the properties, location and amount of cavity absorbent have mainly been studied. For coupled structures, the properties of studs come into play.

The influence of cavity thickness was studied by, e.g. London [3]. Even when a very small cavity between two panels was arranged, there was a considerable improvement in SRI over the single panel. When the thickness of the cavity was increased, the SRI at high frequencies approached a value which was twice that of the single wall. Thus, the second wall was entirely decoupled from the single wall. This occurred in the frequency range 400–1000 Hz, much above the mass-air-mass resonance.

The influence of cavity absorbent was investigated systematically by Loney with steel-studded gypsum walls [4, 5]. It was found that the increase in the SRI as a function of the amount of absorbent was not linear. "The first inch of the absorbent had the greatest effect." With partial cavity treatment, the most useful parameter was the volume of the absorbent. The doubling of volume produces a change of about 2 dB in  $R_w$ . The same rule could be applied for both lateral and thickness dimensions. The density and positioning of the absorbent was of secondary importance.

Ford *et al.* [6] found that if the absorbent locates on the perimeter of the cavity (top, bottom, sides), the SRI is smaller at low frequencies compared to the case when the same quantity of absorbent is located in the middle of the cavity. It was suggested, that because the lowest cavity modes have the maximum particle velocities far from the edges, the middle position of the absorbent leads to better absorption.

According to Narang [7], Novak [8], Au and Byrne [9], and Quirt and Warnock [10], the flow resistivity, rather than the density, or the absorption coefficient, is the appropriate parameter to describe the influence of porous cavity absorbents on SRI. Narang estimated that flow resistivities larger than  $500 \text{ Ns/m}^3$  are not needed to suppress the cavity vibrational modes, usually occurring at middle frequencies. This presupposes that the cavity is almost filled with absorbent. The differences in SRI were less than 5 dB above 250 Hz, when the density varied from 10 to  $130 \text{ kg/m}^3$ , and the flow resistivity varied from 6100 to  $58000 \text{ Ns/m}^3$ . All Narang's experiments were made with

steel studded walls. The effect of the flow resistivity could probably be larger without the studs, when the cavity path is dominant, and smaller with wood studs, when the stud path dominates. Novak studied flow resistivities in the range  $2\,500 \dots 135\,000 \text{ Ns/m}^3$  with uncoupled double walls of cavity spacing 45 mm. The difference was highest at 3 kHz, up to 15 dB, and negligible below 200 Hz.

Von Gösele *et al.* studied the effect of gas fillings in empty cavities of double pane windows [11]. The SRI improved significantly, compared to air-filled cavity, at middle and high frequencies both with heavy and light gases. The improvement of SRI was limited at high frequencies the solid-borne transmission at the edges of the panes so that full effects of the gases could not be determined. The gases had no effect at low frequencies because the double pane glasses used in the experiments were smaller than the wavelength of sound in the cavity.

The effect of rigid studs on the SRI of double walls was studied by Sharp both theoretically and experimentally [12]. The slope of the SRI with uncoupled double wall was 18 dB/octave above the lowest mass-air-mass resonance  $f_0$ . The lowest mass-air-mass resonance, for normal sound incidence angle, is calculated by

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1.8\rho_0 c_0^2}{d} \frac{m'_1 + m'_2}{m'_1 m'_2}}, \quad (1)$$

where  $m'_1$  and  $m'_2$  are the surface masses of the panels ( $\text{kg/m}^2$ ),  $\rho_0 = 1.19 \text{ kg/m}^3$ ,  $c_0 = 343 \text{ m/s}$  and  $d$  is the thickness of the cavity (m). The angle-dependent mass-air-mass resonance is calculated by  $f_0 / \cos \theta$ . Typically, the limiting sound incidence angle is  $\theta = 78^\circ$ . Thus, the mass-air-mass resonance can occur typically in the range  $f_0 \dots 5f_0$  when the cavity is empty. These two frequencies,  $f_0$  and  $5f_0$ , are also called the lower and the upper London frequency, respectively. But when the cavity is sound-absorbing, the sound field in the plane of the panel is so effectively attenuated that the higher London frequency does not exist.

According to Sharp, the slope decreases to 12 dB/octave above a limit frequency  $f_l$  calculated by

$$f_l = \frac{c_0}{6d}. \quad (2)$$

Above this frequency, the coupling between the panels via the airborne path is very weak.

It is possible to build very stiff but also sound insulating double walls using double stud system where both face plates are attached to different studs. However, in several practical constructions, single studs are used. The most usual reasons for using single studs are higher stiffness, thinness, material savings and easier installation. However, the single stud solution considerably decreases the attainable SRI. There is a bridge frequency, above which the slope of the studded wall decreases from 18 dB/octave

to 6 dB/octave. The bridge frequency is calculated for rigid point connections by [12]

$$f_b = f_0 \left( \frac{\pi^3 S}{8 \lambda_c^2 n} \left( \frac{m'_1}{m'_1 + m'_2} \right)^2 \right)^{1/4}, \quad (3)$$

where  $S$  is the area of the wall ( $\text{m}^2$ ),  $n$  is the number of connections on the whole wall area and  $\lambda_c$  is the wavelength at the critical frequency (55.3 mm at 6.3 kHz). The bridge frequency for rigid line connections is

$$f_b = f_0 \left( \frac{\pi b}{2 \lambda_c} \left( \frac{m'_1}{m'_1 + m'_2} \right)^2 \right)^{1/4}, \quad (4)$$

where  $b = S/nl$  is the spacing of vertical studs of length  $l$ . Wood studs can be classified as rigid studs. Sharp studied flexible steel studs only experimentally. They resulted in considerably higher SRI than wood studs. As an alternative to a flexible steel stud, Sharp studied a 6-mm thick, resilient PVC foam tape between the panel and the wood stud. This arrangement gave almost as good results as a flexible steel stud.

Sharp also studied the effect of the thickness and position of the cavity absorbent. A thick perimeter absorbent, e.g. 5...15 cm thick beams, absorbed low-order acoustical modes of the cavity reasonably well. According to the modal coupling theory, the perimeter position should be the optimum location for the absorbent of low-order modes and it is also easy to install. Higher order modes perpendicular to the panel are calculated by

$$f_r = N \frac{c_0}{2d}, \quad (5)$$

where  $N = 1, 2, \dots$ . The lowest higher order mode occurs when  $N = 1$ . The perimeter absorption was no longer effective at and above  $f_r$  and full coverage of panel was needed. The lowest resonance dip with  $N = 1$  could be observed with an empty cavity, even when perimeter absorption was present. Higher resonances ( $N > 1$ ) were no longer distinguishable. At the critical frequency, where the airborne coupling is very important, the perimeter absorption was very ineffective.

In addition, Sharp studied the effect of a  $610 \times 610$  mm non-absorbing lattice in the cavity. It was installed to break the cavity mode pattern to raise the low-order modes. The SRI followed the curve obtained with the sound-absorbing cavity up to 315 Hz band, where the lowest mode of the cavity without the lattice existed. The SRI with the lattice was also considerably higher at high frequencies than without the lattice. This could be interpreted in such a way that cavity absorbents are useless below the lowest cavity resonance where the air acts as a stiffness element. The stiffness depends solely on  $d$  and not on absorption.

Gu and Wang [13] expanded the stud theory of Sharp to flexible studs. The prediction of SRI presupposes the dynamic stiffness of the studs to be determined. The slope of the double wall was 18 dB/octave above the mass-air-mass resonance frequency, both for rigid and flexible studs. The

essential difference between rigid and flexible studs was above the bridge frequency. The slope for flexible studs is 12 dB/octave, while it was 6 dB/octave for wood studs. This implied that flexible studs were superior to rigid studs at high frequencies. Gu and Wang also studied resilient pad strips between the wood stud and the panel. This increased the SRI considerably compared to wood studs alone, but not as much as flexible steel studs.

The bridge frequencies are not equal with rigid and flexible studs. The bridge frequency for flexible point connections is calculated by

$$f_b = \frac{358 \cdot 10^5}{Kd} \sqrt{\frac{S}{n}}, \quad (6)$$

where  $K$  is the lateral equivalent dynamic stiffness (N/m) of the point bridge,  $S$  is panel area and  $n$  is number of point connections. The bridge frequency for flexible line connections are calculated by

$$f_b = \frac{313 \cdot 10^4}{K'd} \sqrt{\frac{S f_c}{nl}}, \quad (7)$$

where  $K'$  is the lateral equivalent stiffness of the stud per unit length  $K'$  (N/m),  $l$  is the length of one stud (m),  $n$  is number of studs and  $f_c$  is the critical frequency of the panel attached to the stud.

Green and Sherry [14, 15, 16] studied the sound insulation of gypsum wallboard partitions in many different forms. In addition to previous references, they found that adhesive attachment (glue) of the panel to the stud decreases the SRI compared to screw attachment. Loosening of the screws did not increase the SRI. But if two or more panels are attached together, adhesive is better than screws running into the studs. As in previous studies, steel studs produced 2...10 dB higher values of STC than wood studs. Resilient channels, where the channel was screwed to the stud and the panel was screwed to the channel, improved the SRI to the same level as with steel studs. With flexible steel studs, the decoupling was sufficient without resilient channels. The effect of cavity thickness was not significant when studs were used. The positive effect of increasing the stud spacing was also very small. They also studied the effect of panel mass in several different ways. Mass symmetric structures gave the same results as asymmetric structures.

Rindel and Hoffmeyer [17] found that a strong resonance phenomenon could occur in double walls with a stud spacing of 300 mm with flexible steel studs. The SRI decreased by 5...15 dB in the range 125...200 Hz compared to stud spacing of 600 mm. Normal resonant modes of 300 mm wide sub-panels, which were formed between two adjacent studs, were suggested to explain this phenomenon. The lowest normal resonance of the panel of width  $L_x$  (m) and height  $L_y$  (m) can be calculated by [18]

$$f_{\min} = \frac{\pi}{2} \sqrt{\frac{B'}{m'}} \left( \frac{m^2}{L_x^2} + \frac{n^2}{L_y^2} \right), \quad n, m = 1, 2, 3, \dots$$

$$B' = \frac{Eh^3}{12(1-\nu^2)}, \quad (8)$$

where  $B'$  (Nm) is the bending stiffness per unit width,  $E$  (Pa) is the Young's modulus,  $\nu$  is the Poisson's ratio,  $h$  (m) is the thickness and  $m'$  (kg/m<sup>2</sup>) is the surface mass of the panel. The lowest resonance occurs with the condition  $n = m = 1$ , which is also the most interesting because it is a monopole radiator and has a very high radiation efficiency. If the panel is mounted on rigid studs, the acoustical width is the distance between studs, which is labelled  $b$  in this study. The edge conditions of the panel (clamped, suspended, free, damping) can considerably modify the predictions of equation (8).

The influence of stud type, stud spacing and screw spacing on the SRI was demonstrated successfully by Quirt and Warnock [10, 19]. Over 250 double-wall experiments were done with gypsum board and also with a thin Lexan plastic having weaker bending stiffness. The panels were attached to the studs by screws. The influence of screw spacing was detrimental for wood studs. Strong resonances could be observed at middle and low frequencies, when the spacing of screws and rigid studs was small, between 200 mm and 400 mm. The resonance shifted to a higher frequency, the more rigidly the studs and screws constrained the gypsum board. The effect was considerably stronger with wood studs than with flexible studs. It was assumed, as by Rindel and Hoffmeyer in equation (8), that a horizontal normal mode, which forms between the studs, was considerably stronger with high screw density than with low density. In practice, screw spacings of 200...400 mm are very common. This result is of great importance because, in laboratory conditions, the screw spacing is usually determined but, in practice, the screw spacing is usually denser than in carefully controlled laboratory tests. Quirt and Warnock agreed with previous studies that the thickness of cavity absorbent was more significant than flow resistivity, absorption coefficient or density. The location of the absorbent appeared to have a small, secondary effect.

The effect of nail spacing in wood-studded plasterboard double panels was studied by Craik and Smith [20]. The decrease in spacing from 150, 300 and 600 mm produced systematically higher values of SRI. It is most common that face plates are attached to the studs using screws. It is natural that the studs are regarded as line connections. However, Craik and Smith showed later that studs should be regarded as line connections when the distance between screws is smaller than a half of the bending wavelength [21]. When the bending wavelength is much smaller than the screw distance, marked by  $b_s$  in this study, each screw connection acts as an independent sound source and its effective radiation area is restricted to a circle of radius  $\lambda_B/2$ . The bending wavelength is calculated by [18]

$$f_B = 2\pi \left( \frac{B'}{m'2\pi f} \right)^{1/4}. \quad (9)$$

According to Cremer and Heckl [22], the radius of the effective radiation area of the panel excited by a point force

(point connection) equals approximately  $\lambda_B/4$ , where  $\lambda_B$  is the wavelength of bending wave in the panel. For line forces, like studs in the cavity exciting the panel, the effective radiation area equals approximately  $\lambda_B/4$ . According to Equation (9), the bending wavelength of a 2-mm thick steel plate is 130 mm at 50 Hz, and 7 mm at 5 kHz. This implies that the radius of the effective radiation area is smaller than the smallest screw spacing used in this study ( $b_s = 170$  mm) at the whole frequency range of interest. This was also pointed out by Craik and Smith when they studied plasterboard with nail spacing of 150 mm [23]. Therefore, the studs should not be modelled as line connections but as line of point connections.

Recently, Muellner and Plotzlin studied the effect of "secondary" wall parameters with double plasterboard wall including the arrangement of screws, the method of screwing, the sealing of the joint between the double wall and the adjoining massive wall and the influence of aging of the sealant [24]. It was found that the deviations are essentially caused by workmanship more than by the changes in quality of the used components. This means that in practical wall installations, statistical variations in workmanship can be so large that the effect of some parameters can not be observed *in situ*.

In addition, extensive product-related studies have been published including double wall structures. The sound insulation of single, double and triple glazing (windows) was studied by Quirt [25, 26]. This investigation clearly showed the influence of glass thickness and interpane spacing on SRI. Cavity resonances in windows were very strong because of negligible cavity absorption. The sound insulation of doors was studied by Hongisto *et al.* [27, 28]. Parametric studies of double structures were only suggestive but it was obvious, on the basis of 18 different steel and wood doors, that the interpanel connections, caused either by cavity absorbent (sandwich-coupling) or coupling via the frames, have a very important role in double-leaf doors. Sound leaks form special problem with windows and doors.

### 3. Materials

This study concentrates on the properties of the cavity, cavity absorbents and studs. The properties of the surface panels are not changed. In all double walls, 2-mm thick steel plates were used as faceplates. The physical properties of steel are given in several references: Poisson's ratio  $\nu = 0.28$ , Young's modulus  $E = 2 \cdot 10^{11}$  N/m<sup>2</sup>, bending stiffness per unit width  $B' = 156$  Nm (with free edge conditions), density  $\rho = 7800$  kg/m<sup>3</sup> and surface mass  $m' = 15.6$  kg/m<sup>2</sup>. The thickness of both panels was  $h = 2$  mm so that the critical frequency, or the lowest coincidence frequency, was  $f_c = 6200$  Hz.

The use of thin steel was motivated for two reasons. Firstly, most previous studies have been done with plasterboard. Its critical frequency is approximately 2500 Hz, that is, the resonance dip occurs in the middle of the frequency range of interest. The coincidence dip makes the

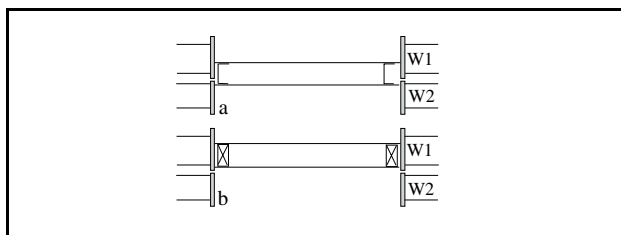


Figure 1. The installation of the specimen into the test opening. a) No studs or flexible studs, b) wood studs.



Figure 2. a) A photograph of the steel studs used in this study. From left: AWS-acoustic wall stud, TC thermo purlin, LR partition stud and LPR partition stud. b) The determination of dynamic stiffness of studs.

observation of, e.g. airborne cavity resonances difficult, because they are often located in the middle frequencies. When 2-mm thick steel is used, the critical frequency is much higher than 3 kHz, that is, both panels follow chiefly the mass law in the frequency range 100...3150 Hz. Secondly, steel keeps its properties during repetitive installations and removals and it is a completely homogeneous product compared to e.g. plasterboard. In addition, steel as a sound insulating constituent was one of main interests in the Finnish project, during which these experiments were carried out.

The size of the steel panels was 1105×2250 mm. The specimen was installed into a test opening of width  $L_x =$

1110 mm, height  $L_y = 2260$  mm and depth 450 mm. The test opening contained two vibration-isolated walls of thickness 250 mm and 200 mm. The faceplates of the double walls were either uncoupled or coupled. The installation was done in two ways (Figure 1). When wood studs were used, both steel panels were installed onto the same wall (W1). But when flexible, 125-mm thick steel studs were used, the steel panels were installed onto separate isolated walls of the test opening to avoid excess coupling via the test opening frame (W1 and W2). Also for uncoupled installations, separate walls were used.

Four different steel studs were used in this study. In all cases, the studs were placed vertically and their horizontal distance was marked by  $b$ . The steel studs are shown in Figure 2 and details are given in Table I.

Three types of wood studs were used. The width was always  $w = 42$  mm. The thicknesses were 45, 84 and 120 mm. The bending stiffnesses  $B$  (Nm) of the wood studs are given in Table I. Seven different mineral wools were used. Their properties are given in Table II. They were manufactured in Finland (Paroc Ltd, Saint-Gobain Isover Ltd). The mineral wool types of this study represent the most typical range of mineral wools used for sound and heat isolation.

## 4. Methods

### 4.1. Sound reduction index

The sound reduction index (SRI) was determined using the pressure method, ISO 140-3 [29]. The source room volume was 81 m<sup>3</sup> and the receiving room volume 113 m<sup>3</sup>. The measurement facility and equipment has been described recently by Hongisto *et al.* [30] so they are not repeated here.

When the pressure method is used, the effect of flanking can be significant, especially when specimens with high sound insulation and small specimen size are measured. Therefore, the maximum achievable SRI of the test arrangement (fixed area, fixed flanking paths) should be known. The test specimen of area  $S$  was covered with a heavy structure so that the direct sound transmission path (Dd) was eliminated. The covering structure consisted of two 4-mm thick steel plates on the perimeters of walls W1 and W2 (see Figure 1) and the cavity was filled by mineral wool. Then, the measurement result was  $R'_T$ . Without the covering structure, the result of the specimen was  $R'_S$ . When  $R'_S < R'_T - 15$  dB, no correction is needed. When  $R'_T - 6$  dB  $< R'_S < R'_T - 15$  dB, the SRI is obtained by  $R_S = -10 \log(10^{-R'_S/10} - 10^{-R'_T/10})$ . When  $R'_S < R'_T - 6$  dB, the correction is constant,  $R_S = R'_S + 1.3$  dB.

In this case, the result is always an underestimate of the true result. Such values were not shown in this study. When flanking was so strong that corrections like above had to be used, the sound intensity method ISO 15186-1 was used [31]. Hongisto *et al.* have recently shown that it is possible to measure even 25 dB higher values of SRI than when using the pressure method, when the intensity method and strong receiving room absorption are used

Table I. The properties of the studs.  $d$  = thickness of the profile,  $w$  = width of the profile,  $t$  = material thickness,  $K'$  = dynamic stiffness per unit length,  $B$  = bending stiffness.

| stud | $d$<br>mm | $w$<br>mm | $t$<br>mm | $K'$<br>MN/m | $B$<br>Nm <sup>2</sup> |
|------|-----------|-----------|-----------|--------------|------------------------|
| AWS  | 125       | 85        | 1.3       | 0.2          | 10400                  |
| TC   | 120       | 52        | 1.6       | 2.8          | 29000                  |
| LR   | 125       | 42        | 1.0       | 3.3          | 20600                  |
| LPR  | 42        | 40        | 0.8       | 0.9          | 3100                   |
| wood | 120       | 42        |           |              | 72000                  |
| wood | 84        | 42        |           |              | 27600                  |
| wood | 42        | 42        |           |              | 3500                   |

Table II. The properties of the cavity absorbers.  $\rho_a$  is the density,  $r$  is the flow resistivity and  $s'$  is the dynamic stiffness. The porosity was between 0.98 and 0.99.

| Name            | $\rho_a$<br>kg/m <sup>3</sup> | $r$<br>kNs/m <sup>3</sup> | $s'$<br>N/m <sup>3</sup> | type       |
|-----------------|-------------------------------|---------------------------|--------------------------|------------|
| TAKU 30 mm      | 63                            | 30                        | 8.9                      | rock wool  |
| TAKU 50 mm      | 57                            | 20                        | 6.3                      | rock wool  |
| PAL 30 mm       | 121                           | 110                       | 14.4                     | rock wool  |
| PAL 50 mm       | 113                           | 70                        | 11.7                     | rock wool  |
| I-KH 30 mm      | 21                            | 8                         |                          | glass wool |
| I-KH 50 mm      | 17                            | 8                         |                          | glass wool |
| EL 100 mm       | 68                            | 40                        | 5.7                      | rock wool  |
| ship wool 15 mm | 216                           | 300                       | 42.7                     | rock wool  |

[30]. In the results, there are occasionally some missing data points at the lowest or highest frequencies. In these frequency bands, the maximum attainable sound reduction index, using the pressure method ( $R_{\max}$ ) or the intensity method ( $R_{I,\max}$ ), was exceeded.

According to literature, the results obtained with intensity method and the pressure method do not deviate more than 2 dB in different frequency bands when the Waterhouse correction is applied [2]. This is valid also in our laboratory. Comparison results were not found necessary to show here.

To estimate the uncertainty of the tests, three quality control tests were done for a 4-mm thick steel plate within six months. In addition, the repeatability value was measured using 6 subsequent tests according to ISO 140-2 [32]. The mounting was the same as that used for the uncoupled steel plates of this study. The same person installed the specimens throughout the study. The repeatability value was 0.1 ... 0.8 dB at 63... 5000 Hz, that is, the acoustical test procedure was very stable. The quality control showed a standard deviation of 0.1 ... 1.2 dB between 63... 5000 Hz and 3.3 dB at 100 Hz (95% confidence interval). This means that the mounting causes some extra uncertainty in the determination of SRI but it is still very small compared to the overall variations in SRI. According to these tests, the results should not contain large uncertainties caused by mounting. The quality control results

were actually much closer to the repeatability values than expected and much smaller than the reproducibility values.

#### 4.2. Total loss factor of the steel panels

The total loss factor  $\eta$  of the steel plates, mounted in the test opening, was determined according to ISO 140-3 Annex E. Total loss factor measurements were made to make the SRI data available also later for, e.g. the validation of prediction models. Most textbooks give only the internal loss factor, typically between 0.0002 and 0.0006. The total loss factor is the sum of internal, radiation and coupling loss factors. The coupling loss depends mostly on the boundary conditions, while the radiation loss depends on the thickness.

The total loss factor was determined by  $\eta = 2.2/fT$  where  $f$  is the frequency and  $T$  is the reverberation time of the plate. The MLS method and time-reversed filtering were used to determine the reverberation time because it is very short for plates in baffle and with rigid boundary conditions [33, 34]. The MLS method was discussed in relation to the measurement of structural damping by Meier and Schmitz [35]. The MLS-method allows a reliable determination of loss factors up to 0.13, while the conventional method using normal filters allows measurements only up to 0.032. Thus, the range is much larger with the modified method. They also pointed out that noise excitation is more reliable than the impulse method based on hammer excitation. With the impulse method, the reverberation time depends on excitation force.

The specimen was installed into the test opening. The MLS signal (Norsonic N840-2), with white noise spectrum, was amplified (Bruel&Kjaer 2707) and the panel was excited at three positions. A 29-cm long push rod, made of steel with a diameter of 10 mm, was used to transmit the vibration from the exciter (Bruel&Kjaer 4805 and 4813) to the specimen. It was pushed lightly against the plate so that the plate did not rattle. The signal-noise ratio of 40 dB was used as a criterion in the MLS measurements. The velocity level of the plate was measured at two positions so that the final result was an average of six reverberation time measurements. Velocity level was measured with a vibration transducer (Bruel&Kjaer 4370) which was attached to the plate using beeswax. The signal was analyzed with a 2-channel real time analyzer (Norsonic N840-2).

Measured total loss factors are shown in Figure 3 for double panels with and without wood studs.

#### 4.3. The properties of cavity absorbers

The flow resistivity, or specific flow resistance per unit thickness, of mineral wools was determined according to ASTM C 522 [36]. The method is essentially the same as ISO 9053 [37] but the ISO standard presupposes considerably smaller flow velocities. The flow resistivity was determined by  $r = SP/Ud_a$ , where  $S$  (m<sup>2</sup>) is the area of the specimen,  $P$  (Pa) is the pressure difference between the front and back side of the specimen,  $U$  (m<sup>3</sup>/s) is the

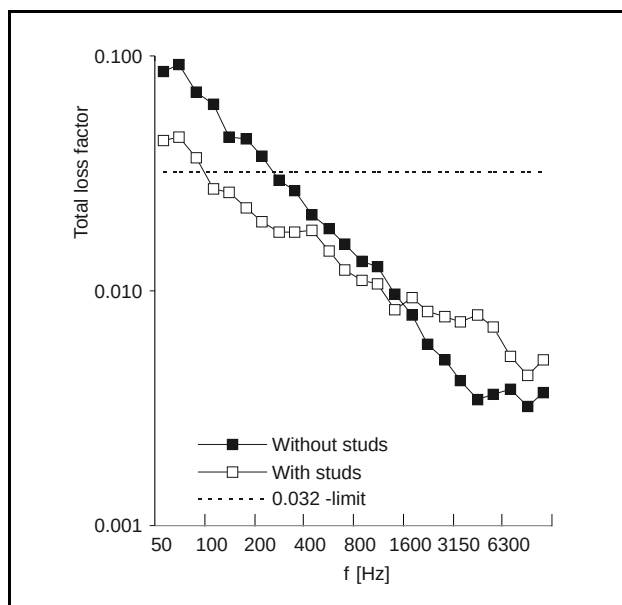


Figure 3. The total loss factor of the 2-mm steel plate in the test opening with studs ( $b = 275$  mm,  $b_s = 170$  mm) and without studs (steel plate alone) measured with MLS-technique. The 0.032-limit curve is the maximum measurable loss factor with the conventional technique.

volume velocity of air flowing through the specimen, and  $d_a$  (m) is the thickness of the specimen.

A plastic circular sewer tube of diameter 104 mm was used as the body. A sample holder was constructed from a similar tube. Airflow in the tube was generated by an air pump (KNF Neuberger). The pressure difference across the sample was measured by a micromanometer (Mikro TT470S), while a rotameter (Kytölä) was used to observe the volume air flow velocity. The volume velocity, 16–26 l/min, through the sample was controlled using an adjustable valve.

The dynamic stiffness per unit area,  $s'$  (N/m<sup>3</sup>), of the mineral wool was measured according to ISO 9052-1 [38]. The values are given in Table I. The stiffness of the absorbents should not play any role in the cavity. However, the data are given to fully characterize the materials.

The porosity of the absorbent, defined by the ratio of the pore volume to total volume, was calculated by  $Y = 1 - \rho_a/\rho_m$  where  $\rho_a$  is the density of the specimen and  $\rho_m$  is the overall density of the mineral wool constituent (2600 kg/m<sup>3</sup>).

#### 4.4. The properties of studs

The dynamic stiffness of flexible studs was determined according to ISO 9052-1 and the definitions given by Gu and Wang for steel studs [13] and by Hopkins *et al.* for rigid wall-ties [39]. The stud fills only a part of the cavity between wall plates, while most of the area is air or filled with absorbent. Therefore, the dynamic stiffness per unit area defined by the standard is not the best descriptor. The definition was changed. The results express the dynamic stiffness per unit length (per one meter), disregarding the

cross-sectional dimensions of the stud. Two studs of total length  $l$  (m) are placed under the load mass  $m_t$  (kg) as shown in Figure 2. The dynamic stiffness per unit length  $K'$  (N/m) is then

$$K' = \frac{4\pi^2 m_t f_r^2}{l}, \quad (10)$$

where  $f_r$  is the resonance frequency of the mass–stud system. The specimen was formed of two studs of length 30 cm. They were placed side by side against the rigid floor. The load mass, with dimensions 300×200 mm and weight 4 kg, was placed on the top of two studs. The load mass was excited with the same apparatus as with the loss factor measurements, but in a vertical direction.

The measurement of stud stiffness was not unambiguous for TC, LR and LPR studs. The back of these studs was straight and much more rigid than that of the AWS stud. It was not absolutely sure that the excitation of the stud corresponded to that occurring in the cavity. However, a beautiful resonance peak could be observed in the FFT spectrum with all studs. Higher resonances were also observed, but they were ignored.

The results are given in Table I. For comparison, Gu and Wang reported that the lateral equivalent stiffness per unit length of their flexible studs was  $K' = 9 \cdot 10^6$  N/m [13]. Craik reported  $K = 1.1 \cdot 10^6 \dots 60 \cdot 10^6$  N/m for 85 mm thick single wall ties used between double masonry walls [18]. Hopkins *et al.* reported  $1.7 \cdot 10^6 \dots 94 \cdot 10^6$  N/m for 50 mm thick wall ties [39]. For comparison, the dynamic stiffness per unit area (N/m<sup>2</sup>) of an airtight air cavity of thickness  $d$  is  $1.4 \cdot 10^5/d$ , that is,  $8 \cdot 10^6$  N/m<sup>2</sup> for a 50 mm thick cavity.

The bending stiffness of studs  $B$  (Nm<sup>2</sup>) were also determined using the static bending experiment. For beams, bending stiffness is defined by  $B = Ewd^3/12$ , where  $E$  (Pa) is the Young's modulus,  $d$  (m) is the height and  $w$  (m) is the width. The stud of length 2.24 m was installed on the top of two supports having a distance of  $L = 2.00$  m. The edge condition of both ends of the stud was free. The stud was loaded by a point force  $F$  in the middle of the stud ( $L/2$ ). The displacement of the stud  $v$  was measured using a dial gauge with a 0.01 mm resolution. Five forces were used. Linear  $F, v$ -curve was fitted to the measurement data which obeyed the relationship  $F(v) = 48BL^{-3}v$ . The bending stiffness  $B$  was determined using linear fitting. For wood studs, the bending stiffness could be easily measured. The measurement of steel studs was problematic because a single steel stud did not keep its shape during loading but bent in a way that does not happen inside the cavity. Therefore, a special arrangement was designed. Two studs were placed side by side on the supports and tied together at 7 points using small steel laths. This arrangement allowed normal bending of the studs. The bending stiffness of this double stud system was divided by two to get the bending stiffness of a single steel stud. The bending stiffnesses of the studs are presented in Table I. They were not used in the analysis but they are reported to give exact data of all materials.



## 5. Results

### 5.1. Overview

In all, 54 sound insulation measurements were made for double panels. All measurement data are not presented here. Only the selected main results and trendlines are presented. The results are given in Figures 4–12. They present the effect of the following parameters:

- cavity thickness for uncoupled structures (Figure 4),
- filling ratio for uncoupled structures (Figure 5),
- stud type and stiffness of studs (Figure 6),
- spacing of wood studs (Figures 7 and 8),
- spacing of steel studs (Figure 9),
- screw spacing between panel and wood studs (Figure 10),
- stud thickness (Figure 11),
- flow resistivity of absorbent (Figure 12).

In addition, the effect of stud spacing for a single panel (Figure 13) was studied to supplement the findings of Figures 6–10.

The structures of Figures 4–13 are shown in Figure 14.

## 6. Discussion

In the following sections, the effect of each parameter is discussed separately. To facilitate the analysis, the values of  $f_0$ ,  $5f_0$ ,  $f_r$  and  $f_l$ , calculated by equations (1), (2) and (5) are given in Table III. The bridge frequencies for flexible line connections are given in Table IV by equation (7).

### 6.1. The influence of cavity thickness in uncoupled structures (Figure 4)

The cavity thickness  $d$  was changed and the cavity was free of studs. In Figure 4a, the cavity is empty. The SRI increases strongly between  $f_0$  and  $f_c$  when  $d$  increases. The slope above  $f_0$ , approximately 10 dB/octave, is quite independent of  $d$ . The lowest higher order cavity resonance  $f_r$  can not be distinguished except when  $d = 250$  mm. However, the dip is quite small. Because steel surface does not absorb sound, it is justified to say that individual high order resonances are of minor importance in practical double wall structures. It is only of academic importance. The lateral field on the plane of panels is more important (lower modes) and it can be effectively damped with absorbent, as shown later in Figure 5.

There was one factor which could weaken the high order modes. The steel plates were slightly convex. The cavity thickness varied by a couple of millimeters. Uncoupled steel panels were clamped at the edges so that the cavity thickness could not be perfectly controlled in the whole area of the specimen. Thus, the quality factor of the resonance is low and very narrow-band resonance can not exist. However, there is no damping in the cavity so resonance should not entirely vanish.

It is interesting that the SRI is independent of  $d$ , approximately 45 dB, at and directly below the critical frequency  $f_c$ . In this frequency range, the vibration of the

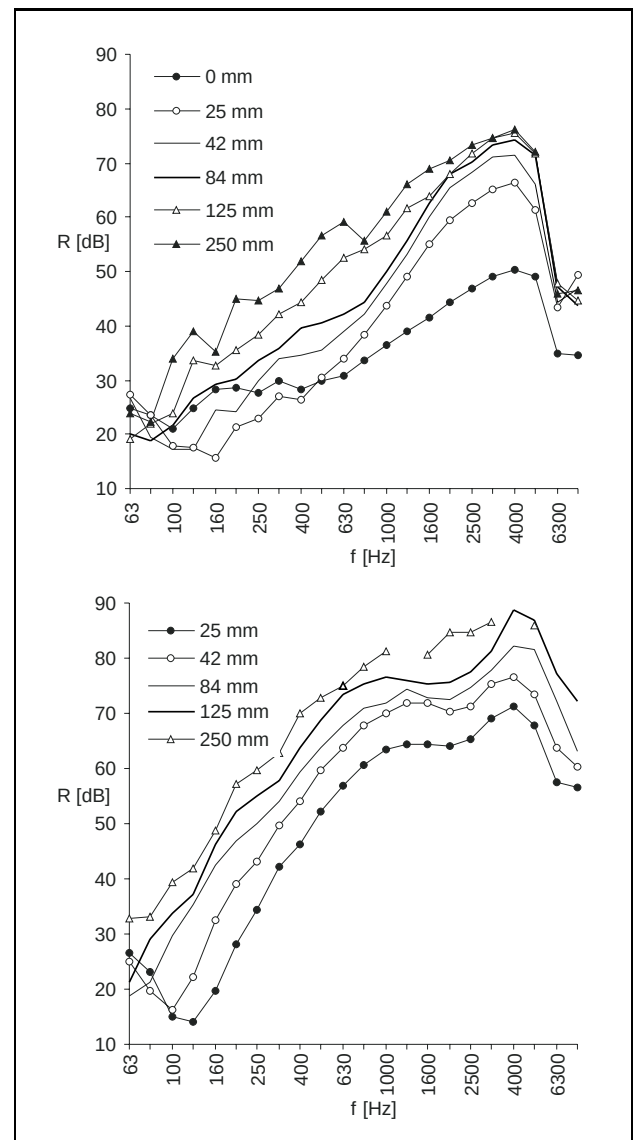


Figure 4. The influence of the cavity thickness  $d$  (mm) in uncoupled structure when the cavity was a) empty or b) absorbing ( $a = 60 \dots 88\%$ ).

panel changes from mass-controlled vibration (forced vibration) to stiffness-controlled vibration (resonant). It is suggested that there is very strong airborne coupling between panels in the coincidence region, which causes the constant SRI, and this coupling does not depend on  $d$  or cavity modes.

When the panels were in contact ( $d = 0$ ), the SRI was better at low frequencies than with cavity walls. The reason is that mass-air-mass resonance does not occur. Above  $f_0$ , this structure was much poorer than cavity walls, as expected.

In Figure 4b, the cavity is absorbing. The SRI is considerably higher than in Figure 4a with same  $d$  values. The mass-air-mass resonance dip at  $f_0$  is narrower than with an empty cavity, that is, the mass-air-mass resonance does not extend to  $5f_0$ , but exists only at  $f_0$ . The slope above  $f_0$  is considerably steeper than with an empty cavity. It was



Table III. The mass-air-mass resonance for normal sound incidence,  $f_0$ , and maximum sound incidence,  $5f_0$ , the first higher order resonance,  $f_r$ , and Sharp's limit frequency,  $f_l$ , for different cavity thicknesses  $d$ .

| $d$ (mm) | $f_0$ (Hz) | $5f_0$ (Hz) | $f_r$ (Hz) | $f_l$ (Hz) |
|----------|------------|-------------|------------|------------|
| 25       | 180        | 867         | 6860       | 2287       |
| 45       | 134        | 646         | 3811       | 1270       |
| 84       | 98         | 473         | 2042       | 681        |
| 125      | 81         | 388         | 1372       | 457        |
| 250      | 57         | 274         | 686        | 229        |

Table IV. The bridge frequencies  $f_b$  (Hz) for steel studs with different stud spacings  $b$  by Equation (7).

| $b$ (m) | LPR  | AWS  | TC  | LR  |
|---------|------|------|-----|-----|
| 1.100   | 4536 | 7348 | 525 | 445 |
| 0.550   | 3703 | 6000 | 429 | 364 |
| 0.275   | 2869 | 4647 | 332 | 282 |

approximately 10 dB/octave with an empty cavity but even 20 dB/octave or higher with an absorbing cavity. Surprisingly, there is a plateau at 800–2000 Hz above which there is a new rise below the critical frequency. The plateau is more than two octaves below the critical frequency so that it can not be explained by sub-critical resonant vibration. The lower end of the plateau seems to decrease slightly with increasing  $d$ . The position of the plateau could not be explained by lowest high-order resonance  $f_r$ , because it decreases from 2.3 kHz to 230 Hz when  $d$  increases from 25 mm to 250 mm. Flanking can not be the reason for the plateau because the level of the plateau is not constant.

Nor can the niche-effect explain the plateau, which is independent of  $d$ . The niche effect should be stronger with larger niche depth (or smaller  $d$ ). E.g. Cops *et al.* [40] obtained at most 5 dB better results with extreme position of the specimen (niche depths were 0 and 70 cm) than with central position of specimen (equal niche depths). Guy and Sauer [41] and Cops and Minten [42] had similar results. The reason for low SRI in the middle position was that the airborne modes of the niche on both sides of the specimen are effectively coupled while in the edge position there is no coupling. Differences existed between the lowest airborne resonance of the test opening and the critical frequency of the panel. The niche effect was minimized by avoiding edge positioning of the specimen. The niche ratio 1:2 is prescribed in ISO 140-3. In our study, the specimen was always placed as centrally as possible so that the asymmetry of niches was not significant. Thus, the origin of the plateau remains unknown.

It is also notable that the depth of the coincidence dip was 30 dB when the cavity was empty, but only 15 dB when it was absorbing. The SRI also increased at  $f_c$  when  $d$  increased, which did not happen with an empty cavity. As mentioned above, there is strong airborne coupling between the panels at  $f_c$  and the cavity modes play a

very small role in transmission. But when the absorbent is present, sound dissipates during the first passage through the absorbent, especially at high frequencies. This will be demonstrated below. Therefore, the increase in SRI with increasing  $d$  is possible at  $f_c$  only with an absorbing cavity.

## 6.2. The influence of the filling ratio of absorbent in uncoupled structures (Figure 5)

The filling ratio describes the relative volume of absorbent in the cavity. It is defined by

$$a = 100 \frac{d_a}{d} \quad \%, \quad (11)$$

where  $d$  and  $d_a$  are the thicknesses of the cavity and absorbent, respectively. The filling ratio was changed when the cavity was free of studs. Measurements were carried out with five different cavity spacings, viz.  $d = 25, 42, 84, 125$  and  $250$  mm, of which two, 42 mm and 125 mm, are shown in Figures 5a and 5b. The trendlines were the same for every  $d$  so that full understanding can be achieved by these two figures.

The SRI increases strongly even with small increments of absorbent presupposed its amount is initially small. The increment from 0% to 24% gives a much larger increase in SRI than the increment from 24% to 88%. This agrees with previous studies [4, 5]. The difference in SRI between the full and empty cavity did not depend on cavity thickness. The increase in SRI was largest, 20 ... 30 dB, between 500 and 1000 Hz and above  $f_c$ , and smallest at and below  $f_0$ . At the plateau 800–2500 Hz, all curves were very close to each other.

From Figure 5a it is quite obvious that the value of  $f_0$  slightly decreases when absorbent is added to the cavity. This agrees with the theory of Narang [7]. Sound-absorbing material makes the sound propagation at low frequencies an isothermal process, which enlarges the effective size of the cavity. The speed of sound increases with the mineral wool by a factor of  $\sqrt{\gamma} = 1.183$ , where  $\gamma$  is the ratio of specific heats at constant pressure to those at constant volume for air. Thus, the mass-air-mass resonance frequency is multiplied by a factor of  $1.4^{-0.5} = 0.845$ , which means nearly one third octave reduction.

The increment of SRI with increasing absorption was negligible at and below  $f_0$ . According to Fahy, the SRI should increase also at  $f_0$  if the attenuation during one passage of sound is high enough [43]. The attenuation can be estimated in terms of the propagation constant, which is  $\gamma = \alpha + j\beta$ . Here,  $\alpha$  is the real part, which is often called the attenuation constant. Sufficient attenuation occurs when  $\alpha d \approx rd/2\rho_0 c_0 > 1$  because the pressure amplitude is reduced by at least a factor of three during one passage through the material. In that case, an effective airborne decoupling should occur between the panels [43]. In Figure 5a, the flow resistivity was  $r = 8000 \text{ Ns/m}^3$  so that  $\alpha d = 0.42 < 1$ . Thus, the theory of Fahy was valid. In Figure 5b,  $\alpha d = 1.25 > 1$ , but the validation of the theory could not be made reliably because  $f_0$  was below 80 Hz.

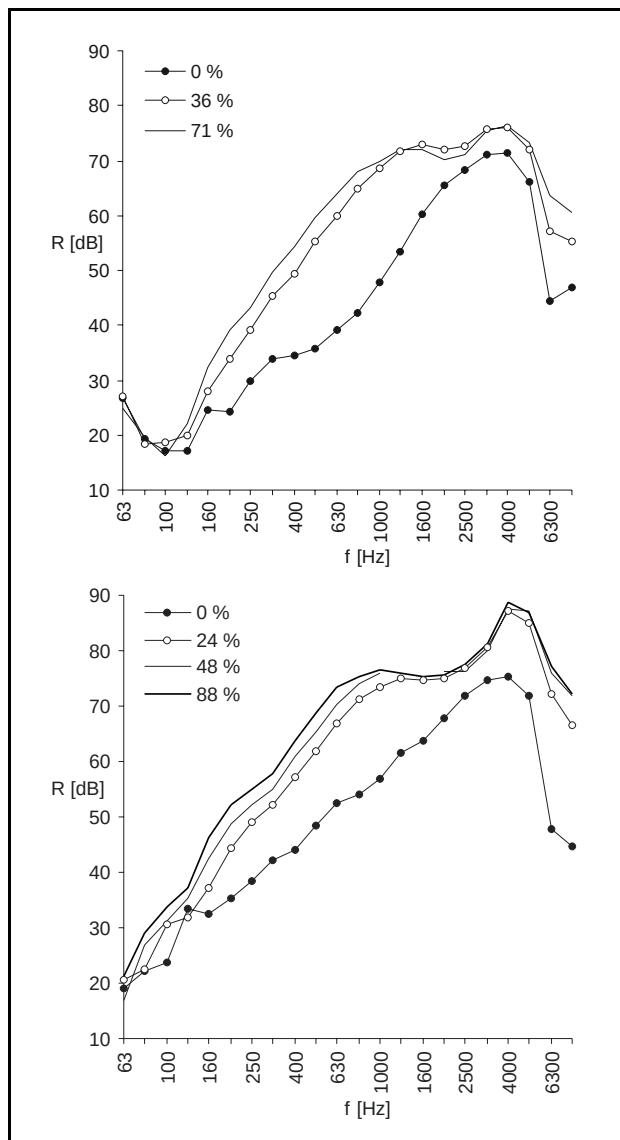


Figure 5. The influence of filling ratio  $a$  (%) in uncoupled structure. The thickness of the cavity was a)  $d = 42$  mm and b)  $d = 125$  mm.

If more resistive absorbent were used in Figure 5a, an increase in SRI at  $f_0$  with increasing absorption could have been observed.

It is notable that the slope of SRI is even 20 dB/octave with a large filling ratio and just above  $f_0$ . However, all curves, with any non-zero filling ratio, end in the same plateau at approximately 1250 Hz. The shape of the plateau was already discussed in Figure 4b. At the plateau, the filling ratio makes no difference, provided it deviates from zero. The same behaviour was observed also with  $d = 25, 84$  and  $250$  mm, which are not shown here. Mulholland *et al.* [44] presupposed that the curves would come together above the highest mass-air-mass resonance frequency, or  $5f_0$  (see equation 2). The values of  $5f_0$  are calculated in Table III. They should reduce with increasing  $d$  but the position of the plateau was constant. Thus, the plateau can not be explained by  $5f_0$ . There is a possibility that the plateau is caused by flanking via path  $F_d$ , that is,

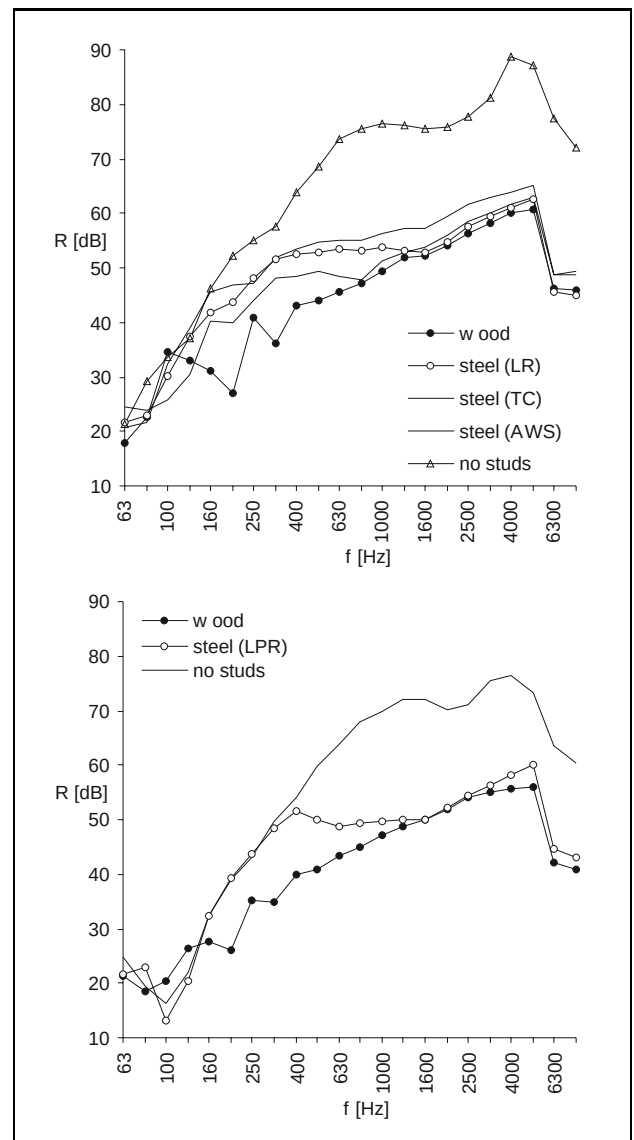


Figure 6. The influence of the stud type (steel, wood, no studs). The stud spacing was  $b = 550$  mm and the cavity spacing was a)  $d = 120/125$  mm and b)  $d = 42/45$  mm. The cavity was absorbing.

the flanking path from the source room walls to the specimen. Flanking is possible because the SRI is constant at the plateau. However, the plateau does not affect the main conclusions of Figure 5 so that further analysis was found unnecessary.

### 6.3. The influence of stud type (Figure 6)

The stud type was changed when the cavity was absorbing and  $d = 120/125$  mm (Figure 6a) or  $d = 42$  mm (Figure 6b). Figure 6a is dealt with first. There were 4 different commercial studs of thickness 125 mm. It is shown below that the effect of  $d$  in studded walls is small. Therefore, both stud thicknesses,  $d = 120$  mm and  $125$  mm, were given in the same graph.

It was expected that the dynamic stiffness of the stud would be the most important factor explaining the differ-

ences of SRI between different studs. The measured dynamic stiffnesses are given in Table I. The AWS stud had a ten times smaller stiffness than the others. This implies that it should have the best sound insulation performance. The reason for the lower dynamic stiffness becomes evident in Figure 2. The shape of the AWS stud is designed to give small dynamic stiffness but reasonable bending stiffness. The LPR stud, which was 42 mm thick, had the same dynamic stiffness as the AWS stud. This was also expected because the LPR stud is manufactured of considerably thinner steel.

According to measurements, the AWS stud gave the best performance. The difference between the AWS stud and wood stud was always larger than 5 dB between 125 and 2000 Hz. Above 2 kHz, the difference was smaller, 3 dB. The results with all studs except the AWS, lie within 2 dB above 1250 Hz. It seems that the type of stud is important at low and middle frequencies but not at higher frequencies. This did not agree with the theory of Gu and Wang, who suggested that the effect of dynamic stiffness is most important above the bridge frequency [13].

It is obvious that the AWS-stud has the smallest stiffness and also the highest SRI. However, although the stiffnesses of LR and TC studs are almost equal, the LR stud results in higher SRI. This disagreement could also be explained by our measurement method of  $K'$ : it is uncertain whether this fully corresponded to the situation inside the cavity (see Methods).

The predicted bridge frequencies  $f_b$  for steel studs are given in Table IV. The values of  $f_b$  increase with decreasing dynamic stiffness of the stud. No correlation can be found between predicted  $f_b$  and measurements. The bridge frequency seems to be located in a very wide range, depending on  $K'$ , and is typically much higher than that of rigid studs, around 200 Hz. The theory of Gu and Wang predicts a slope of 12 dB/octave above the bridge frequency. For TC and LR studs, the bridge frequencies are close to 400 Hz. However, no such slope was observed. The AWS stud gave the same results as the uncoupled double wall up to 160 Hz, with a slope of 24 dB/octave. Above 160 Hz, the slope becomes much smaller, 3...5 dB/octave, and this slope continues up to the critical frequency.

The influence of stud type was studied also with  $d = 42/45$  mm in Figure 6b. The steel stud resulted in a 6 dB lower SRI at  $f_0$  than the wood stud, but above 125 Hz, the steel stud was better. Again, as in Figure 6a, the curves come together at 1250 Hz, independent of stud type. The steel stud gave the same results with the uncoupled double wall up to 400 Hz. In Figure 6a, the AWS stud did the same only up to 160 Hz, although its dynamic stiffness was considerably smaller than that of LPR stud. This is also contrary to the theory of Gu and Wang.

#### 6.4. The influence of stud spacing and cavity thickness with wood studs (Figure 7)

The stud spacing of rigid wood studs was changed when the cavity was absorbing. The cavity spacings were  $d = 45$  mm (Figure 7a) and  $d = 120$  mm (Figure 7b). Exper-

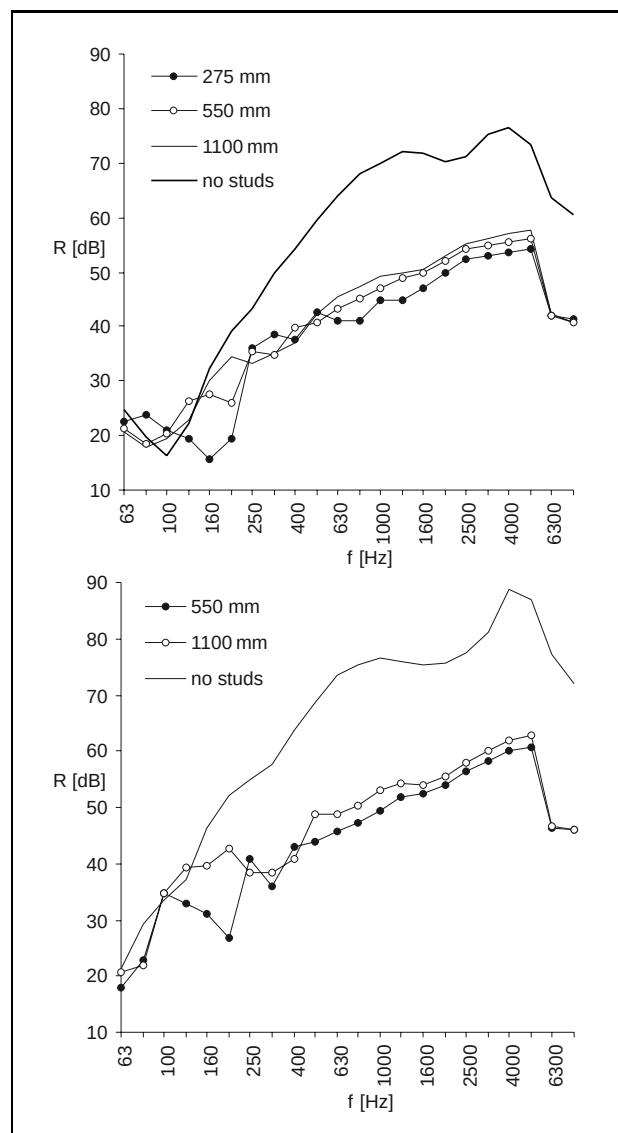


Figure 7. The influence of stud spacing  $b$  (mm) with wood studs. The cavity was absorbing and the screw spacing was  $b_s = 170$  mm. The thickness of the cavity was a)  $d = 45$  mm and b)  $d = 120$  mm.

iments with  $d = 84$  mm resulted in similar conclusions (Figure 8a). Vertical studs were mounted with three spacings, viz.  $b = 275, 550$  and  $1100$  mm. This range covers the most typical stud spacings *in situ*. The largest value,  $b = 1100$  mm, corresponds to the width of the test opening.

The effect of  $b$  on SRI at middle and high frequencies was smaller than expected according to the theory of Sharp [11]. The SRI increased with increasing  $b$ , but the increment was only 1...5 dB above the bridge frequency. According to Sharp's theory, the increment should be 6 dB. In addition, Sharp's theory predicts that the slope should be reduced from 18 dB/octave to 6 dB/octave above the bridge frequency. Here, the slope 18 dB/octave below the bridge frequency could not be observed at all. Above 200 Hz, the slope was approximately 4 dB/octave.

The bridge frequency  $f_b$  was calculated by Equation (3) for point connections. It was shown previously by Craik

and Smith [23] that wood studs, against which the face plates are connected by screws, should be treated as a series of point connections rather than line connections. This is because half of the bending wavelength in the panel is considerably smaller than the distance between point connections, or  $b_s$ . In Figure 7, we had  $b_s = 170$  mm, so the total number of point connections in the specimen was  $n = 42$ . From Equation (3) we get  $f_b = 280$  Hz when  $d = 45$  mm, and  $f_b = 170$  Hz when  $d = 125$  mm. The low frequency range is dominated by resonances so that  $f_b$  can not be absolutely distinguished, but the predicted  $f_b$  match quite well with the measured data. The constant slope of 4 dB/octave seems to start above the bridge frequency so the theory behind Equation (3) is valid.

Below  $f_0$ , the SRI of the uncoupled wall was lower than with the studded wall (Figure 7a). The stiffening of the plate apparently weakens the mass-air-mass resonance. The resonance is caused by asymmetric motion of the plates and is partially prevented by studs.

There was a strong resonance, even 18 dB, at 160 Hz when  $b = 275$  mm. The resonance diminished with increasing  $b$ . It has been suggested in the literature that the studs divide the face plate into sections, or sub-panels, that have the same height as the specimen,  $L_y$ , but the width of the panel,  $L_x$ , is replaced by the stud spacing  $b$ . This should result in new normal modes with a higher frequency than those predicted by using the whole panel dimensions in equation (8) [10, 17]. However, the dips at 160...315 Hz do not match the lowest normal modes of sub-panels. The predicted normal modes are considerably lower for every  $b$ . E.g. for 2-mm thick steel, the lowest normal modes are  $f_{11} = 93$  Hz, 20 Hz and 5 Hz for  $b = 275$  mm, 550 mm and 1100 mm, respectively. In the calculation of  $f_{11}$ , we used  $L_x = b - 42$  mm because the width of the stud was  $w = 42$  mm and the width of the sub-panel is 42 mm smaller than the stud spacing. It will be shown later that studded single steel panels give the same resonance frequencies as in Figure 7 for double walls.

It is reasonable to assume that the boundary condition of the panel at the stud is not free. The wood studs were very stiff and the panel was screwed against them using dense screw spacing. According to the basic static theory, the bending stiffness is  $B = FL^3/48\nu$  for free boundary conditions but  $B = FL^3/192\nu$  for fixed boundary conditions, that is, fourfold. When this is applied in Equation (8), the normal modes are  $f_{11} = 185$  Hz, 40 Hz and 11 Hz for  $b = 275$  mm, 550 mm and 1100 mm, respectively. In the present study, the predicted value  $f_{11} = 185$  Hz matched very well with measurements. Thus, it can be assumed that the sub-panel model is valid.

There is also another possible explanation for the resonance dips. It is obvious that the vertical bending stiffness of the double wall increases considerably because of the wood studs. This can also lead to vertical normal modes, called the global normal mode of the double wall. Wave propagation can be controlled by whole-section bending stiffness at low frequencies, while at high frequencies, it is controlled by individual face plate stiffnesses.

To investigate this hypothesis, the vertical bending stiffness was determined for a whole double wall. The measurement method was essentially the same as that used to determine the bending stiffness of studs (see Methods). However, to calculate the normal mode by Equation (8), the bending stiffness  $B'$  (Nm) per unit width of the plate is needed. This was calculated by  $B' = B/w$ , where  $w$  is the width of the double wall, or  $w = 1.105$  m. The values of  $B'$  were determined when  $d = 42$  mm,  $b_s = 170$  mm and  $b = 275$  mm. These values are the same as in the lowest curve of Figure 7a. The result was  $B' = 77000$  Nm, while it was  $B' = 156$  Nm for 2 mm thick steel plate. Thus, the studded double wall is 500 times stiffer than the faceplate. Using the measured value of  $B'$  of the studded double wall, the whole-section normal mode was calculated by equation (8). Because the lowest vertical normal mode was investigated, we used  $m = 0$ ,  $n = 1$ ,  $L_y = 2.25$  m and the surface mass of the whole wall,  $m' = 32$  kg/m<sup>2</sup>. The result was  $f_{11} = 15$  Hz (free edge condition). This means that the hypothesis of the whole-section normal mode was probably invalid. Nor does the whole-section normal mode cause the low frequency resonance in Figure 7 or in Figures 8 and 10a.

In general, the effect of stud spacing at middle and high frequencies was much smaller than expected, but larger at low frequencies. A better understanding of low frequency resonances is necessary to be able to design optimal studs and stud spacings for multilayer walls. This presupposes more measurements and physical analyses.

### 6.5. The influence of stud spacing and screw spacing with wood studs (Figure 8)

The stud spacing of rigid wood studs was changed when the cavity was absorbing. The cavity thickness was  $d = 84$  mm and the screw spacing was  $b_s = 170$  mm (Figure 8a) and 680 mm (Figure 8b). The measurements were performed also with  $b_s = 340$  mm. The result needed not to be shown because of similar findings as here.

The influence of stud spacing is most important below 250 Hz where the above-mentioned low-frequency resonances occurred. When the screw spacing was small, the resonance was strongest. Above 250 Hz, the influence of stud spacing was less than 3 dB. The slope is approximately 5 dB/octave. All three curves with large screw spacing are 10 dB higher above 250 Hz than with small screw spacing.

It is obvious that when wood studs are used, it is important to use the widest possible spacing between screws and studs. A double stud system would, naturally, be the best choice so that no mechanical connections exist.

### 6.6. The influence of stud spacing with flexible steel studs (Figure 9)

The stud spacing of flexible steel studs (AWS) was changed when the cavity was absorbing and the cavity thickness was  $d = 125$  mm. The resonance occurs at the same frequency, 200 Hz, as with wood studs and with the

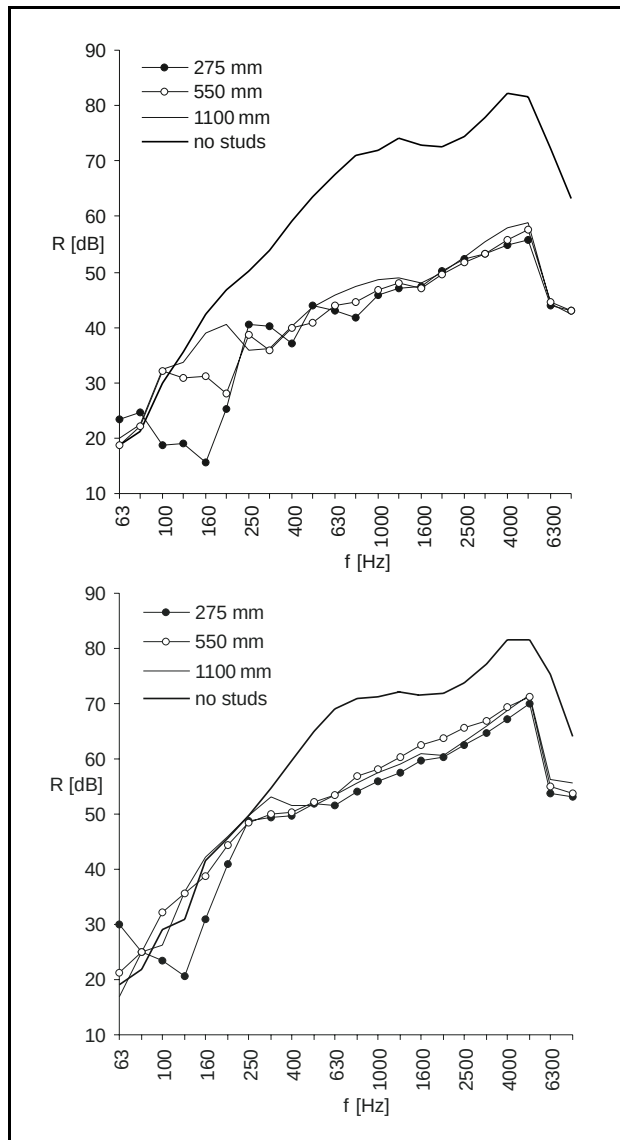


Figure 8. The influence of stud spacing  $b$  (m) with wood studs. The cavity was absorbing and the cavity thickness was  $d = 84$  mm. The screw spacing was a)  $b_s = 170$  mm and b)  $b_s = 680$  mm.

smallest stud spacing, but the resonance is considerably weaker. The resonance dip is only 10 dB when it was as much as 20 dB with wood studs. Because the bending stiffness of the 125 mm thick wood stud is considerably larger than that of the AWS stud, the whole-section normal mode model discussed in Figure 7 is invalid here too. The independency of the resonance frequency on the stud stiffness indicates that the sub-panel normal mode theory seems to be the best explanation. The reason for the weak resonance could be the flexibility of steel studs. The formation of sub-panels of width  $b$  is less dramatic. The only problem with the sub-panel theory is that the resonance frequency should decrease with increasing stud spacing, but it does not, at least in Figures 7–9. Only the amplitude of the resonance decreases with increasing stud spacing.

The influence of stud spacing above 200 Hz was roughly the same as with rigid studs, viz. only a couple of decibels.

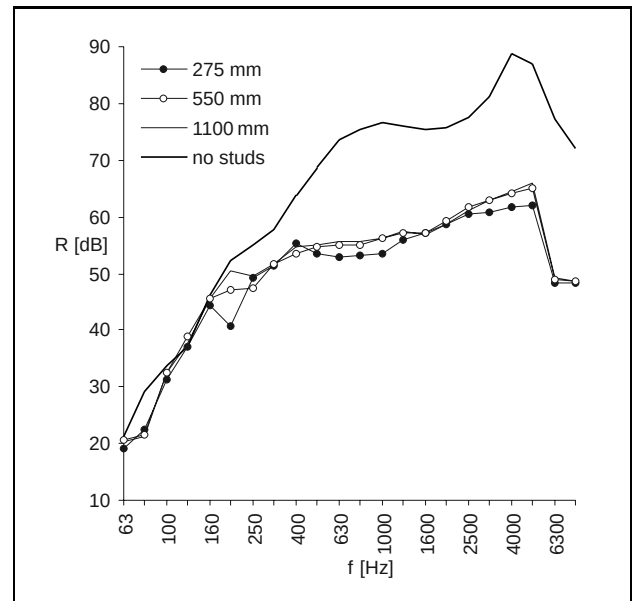


Figure 9. The influence of stud spacing  $b$  (mm) with flexible steel studs (AWS type). The cavity spacing was  $d = 125$  mm and absorbing. The screw spacing was  $b_s = 170$  mm.

The overall effect of stud spacing was practically negligible with flexible studs compared to rigid studs.

#### 6.7. The influence of screw spacing with wood studs (Figure 10)

The screw spacing  $b_s$  was changed when the wood stud spacing was  $b = 550$  mm. The cavity was absorbing and its thickness was 84 mm. The effect of screw spacing was considerably stronger than the effect of stud spacing in Figures 7–9. The screw spacing affected the amplitude of the low-frequency resonance. The normal mode was less prominent when the number of screws was reduced. The influence of screw spacing was not limited to the low frequency range, as happened with stud spacing, but the effect was strongest, and almost linear, at middle and high frequencies.

In Figures 7–9, the SRI at  $f_c$  was independent of stud type and stud spacing. It was expounded that the airborne coupling between the plates was strong. However, the same phenomenon did not occur with varying screw spacing. The reason for this remained unknown. The results clearly implied that every screw acts as a strong transmission path also in the coincidence region. The influence of screw spacing was also studied with the face plate alone. The test arrangement was identical with that in Figure 10a but the other face plate and mineral wool were removed. The results are not presented here because no evidence of screw spacing was found. This implies that the screw spacing does not affect the panel modes.

#### 6.8. The influence of stud thickness (Figure 11)

The thickness of the wood studs was changed. The cavity was absorbing. In Figure 11a, the stud spacing was

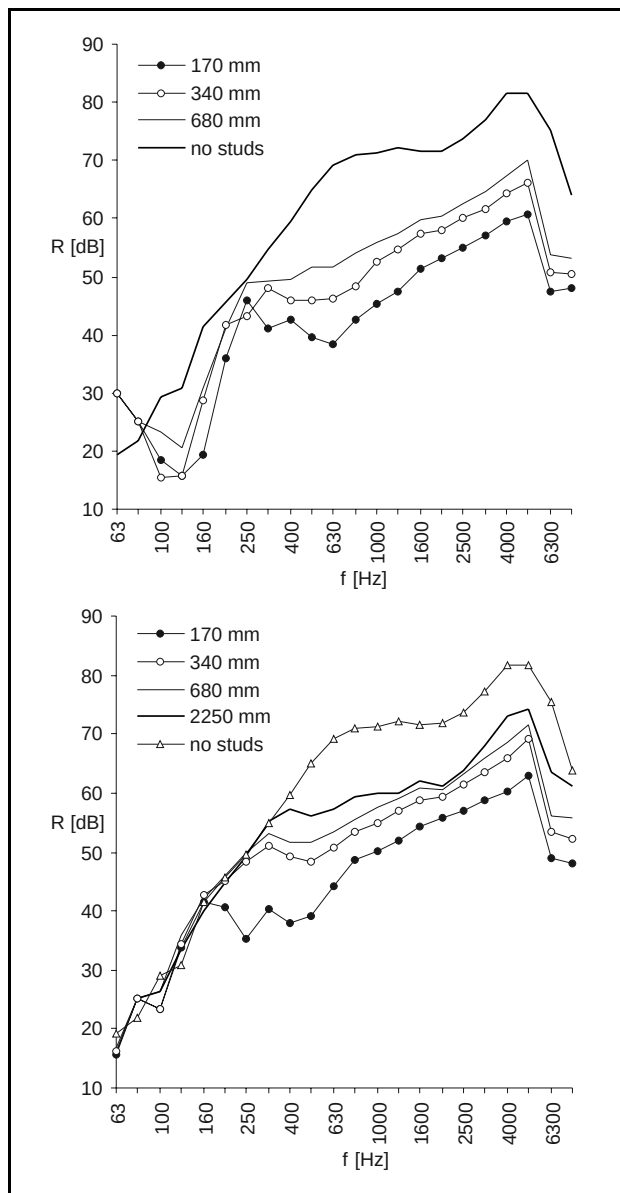


Figure 10. The influence of screw spacing  $b_s$  (mm) with wood studs. The cavity thickness was  $d = 84$  mm and the cavity was absorbing. The stud spacing was a)  $b = 275$  mm and b)  $b = 1100$  mm.

$b = 550$  mm. Above 400 Hz, only a small increase with increasing stud thickness could be found. The reason for the small difference was that the coupling via air is weak when rigid studs are used. When the main transmission path is the stud, its thickness is rather insignificant. The largest differences occurred at low frequencies, because below  $f_b$ , the main transmission path was the air cavity. For rigid point connections with  $b_s = 170$  mm, the bridge frequency is  $f_b = 280$  Hz for  $d = 45$  mm, and 170 Hz for  $d = 120$  mm. It was already shown in Figure 4 that the effect of cavity thickness was very important for the cavity-controlled region. However, at 200 and 315 Hz, the SRI was constant. The reason is probably the normal modes of the panel sections between the studs, as discussed above. These are not affected by the changes in the cavity.

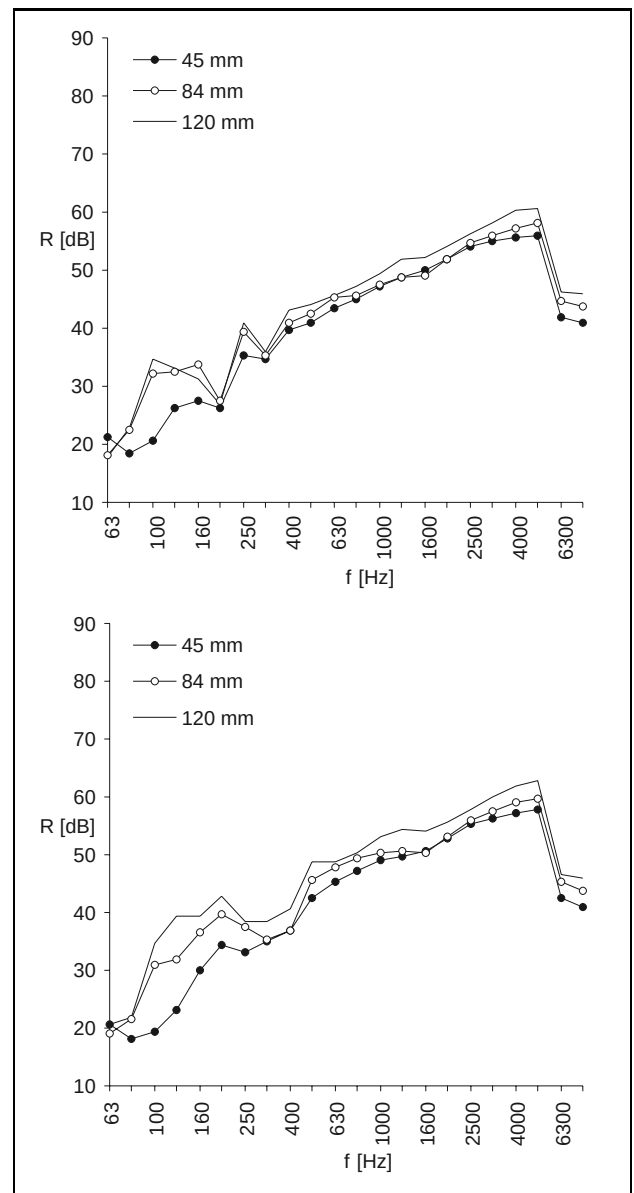


Figure 11. The influence of cavity thickness  $d$  (mm) with wood studs. The stud spacing was a)  $b = 550$  mm and b)  $b = 1100$  mm. The cavity was absorbing and the screw spacing was  $b_s = 170$  mm.

In Figure 11b, the stud spacing was  $b = 1100$  mm. In this case, the influence of the airborne path is more important because the number of studs was smaller. This can be seen especially at low frequencies where SRI is higher. At high frequencies, the increase of SRI with increasing  $d$  was of the same order, 2...3 dB, both in Figures 11a and 11b.

Unfortunately, the effect of cavity spacing with flexible studs could not be investigated because of the lack of commercial studs with varying thickness but constant dynamic stiffness. On the basis of previous figures, the influence of cavity spacing should be larger with flexible studs than with wood studs. The stud path is less important with flexible studs and, therefore, the dependence on the stud thickness is expected to be closer to the findings of Figure 4.



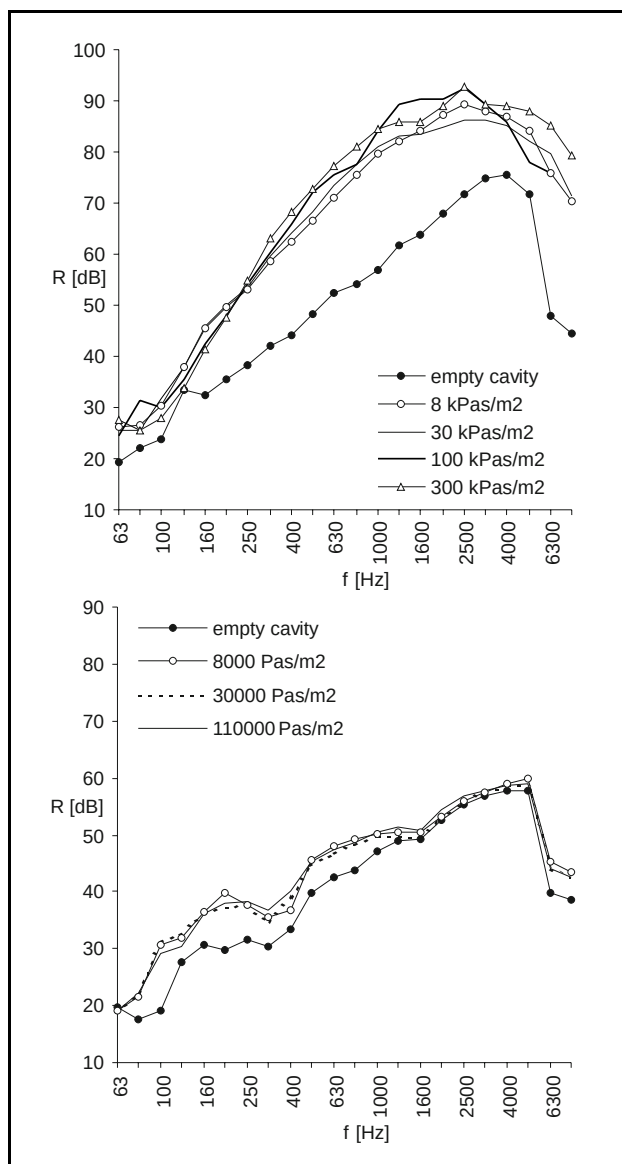


Figure 12. The influence of flow resistivity  $r'$  a) without studs ( $d = 125$  mm) and b) with wood studs having  $b = 1100$  mm and  $b_s = 170$  mm ( $d = 84$  mm).

### 6.9. The influence of flow resistivity and density of cavity absorbent (Figure 12)

The flow resistivity  $r'$  of the cavity absorbent was changed in two cases: without interpanel connections (Figure 12a) and with wood studs (Figure 12b). The values of  $r'$  were 8...300 kPa s/m<sup>2</sup>. Without studs, the effect of flow resistivity was, at the most, 6 dB. This was less than expected on the basis of the literature. It is possible that the range of flow resistivities was not wide enough to find larger differences. All materials were essentially the same mineral wool. It would be interesting to repeat the study using also different materials and an even larger range of flow resistivity. Especially values below 5 kPa s/m<sup>2</sup> would be interesting. However, mineral wool absorbents are so cheap that they are mostly used whenever cavity absorption is needed. The range of flow resistivities of commercial min-

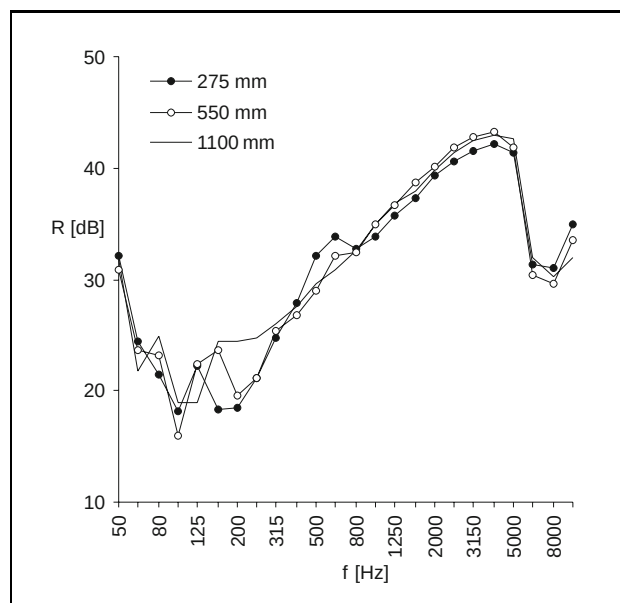


Figure 13. The influence of stud spacing  $b$  on a 2-mm thick steel plate. The screw spacing was  $b_s = 170$  mm.

eral wools was fully covered in Figure 12 so it is probably not necessary to study the effect of poorer absorbents. The results would definitely be somewhere between the curves  $r' = 0$  and 8000 Pa s/m<sup>2</sup>.

With studs, the effect of flow resistivity was smaller, of the same order as the measurement uncertainty, 0...2 dB (Figure 12b). The stud path was so dominant that the changes in the flow resistivity could not be distinguished. However, there was an obvious difference between 0 and 8000 Pa s/m<sup>2</sup>, especially. Thus, it is not completely useless to place absorbents in the studded cavity. It is also notable that the absorbent has the strongest effect on low and middle frequencies, where the cavity path plays an important role.

It should be noted that the surface mass of the heaviest mineral wool was 22 kg/m<sup>2</sup>, that is, 40% of the total mass of the wall. However, there is no evidence that the mass increase had an effect on SRI, which could be expected by the mass law. This point needs further research because dense mineral wool (also called ship wool) is often used to replace, e.g. building panels in multilayer structures because of its low price and good fire protection.

### 6.10. Investigations with a single steel plate (Figure 13)

The effect of wood stud spacing,  $b$ , on a single 2-mm thick steel panel was studied to gain more insight into the low frequency resonance(s) observed with dense wood studs. In Figure 13, there is a dip at 200 Hz with  $b = 275$  mm, and the resonance diminishes with increasing  $b$ . These findings agree with the resonances observed in Figures 7, 8 and 10. Thus, the studded steel panel, and the sub-panel normal mode, should be the origin of the resonance. The resonance is stronger with the double panel because there are two identical panel resonators.

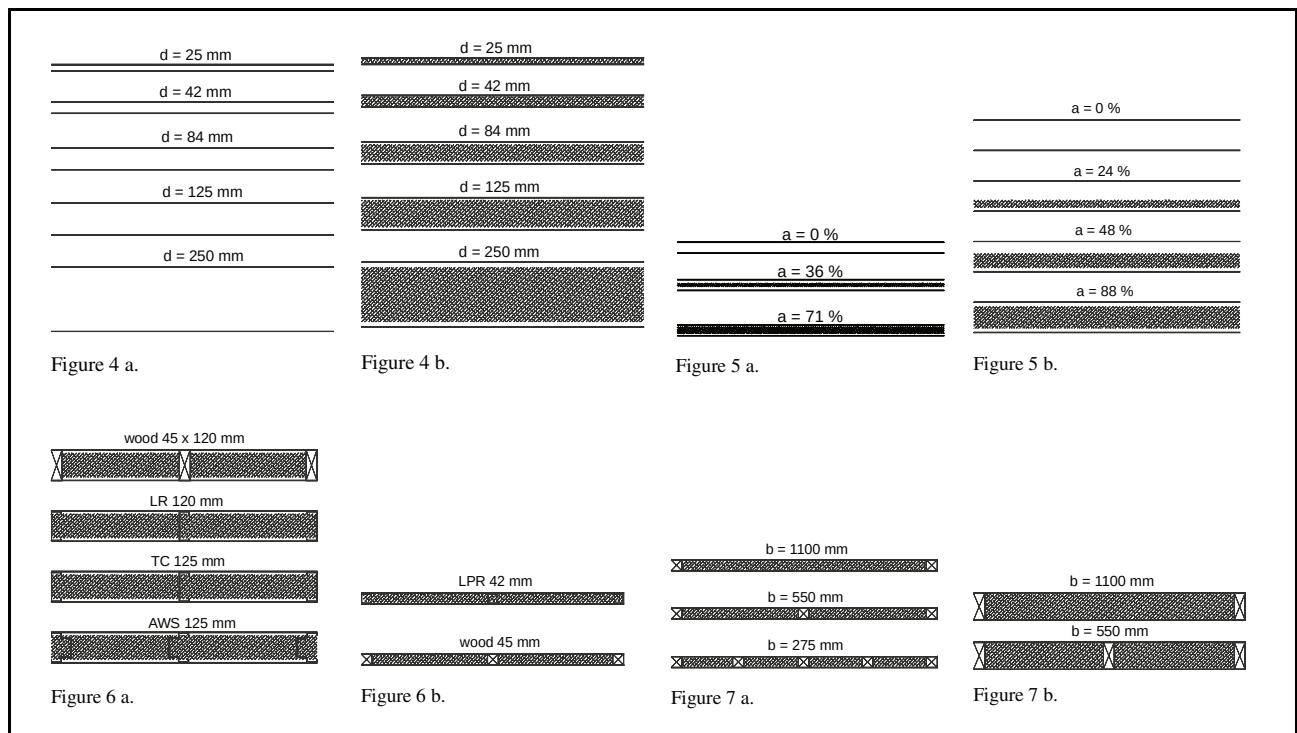


Figure 14. Schematic structure drawings of the double wall sets used in Figures 4–13.

Variations of SRI at 500–630 Hz could not be explained but they are negligible.

The effect of screw spacing was also studied with single panels using three values,  $b_s = 170, 340$  and  $680$  mm. The stud spacing was  $b=550$  mm. The screw spacing did not affect the SRI. The results are not shown here because the curves were all within 1 dB.

### 6.11. General discussion

The main point of this study was to give an overall view of the most important double wall parameters. The purpose was to concentrate on cavity parameters. Certain double wall parameters were not studied here to reduce the number of tests and to reduce the length of the paper. The explanations follow.

The position of the absorbent was not studied. Firstly, previous studies have shown that the position of the absorbent has only a minor secondary influence. Secondly, practical structures seldom allow for other installations than full area coverage of absorbent. However, the thickness of the absorbent is an important parameter.

The influence of surface mass of the panel has been studied in several previous papers and it is probably best known. To reduce the number of tests, the surface panels were not changed in the study. The panel mass ( $15.6 \text{ kg/m}^2$ ) represents a typical situation in lightweight double walls. However, constant mass is an obvious shortcoming of this study and in further studies it should be considered.

Nor was the stiffness of the panel studied. Steel is considerably looser than, e.g. plasterboard with the same sur-

face mass. This results in a higher critical frequency. It would be very interesting to repeat the same experiments with a panel having a considerably lower critical frequency to see how the SRI behaves for pure resonant vibration above the critical frequency. The effect of studs above the coincidence was not studied here at all. This also motivates further studies with stiffer materials.

The specimen size was continuously  $2.5 \text{ m}^2$  ( $1100 \times 2260 \text{ mm}$ ) in this study. ISO 140-3 presupposes a size of approximately  $10 \text{ m}^2$  for partition walls. However, smaller size was motivated because it was not the intention to find the absolute values to be used, e.g., in commercial applications, but to find the effect of different parameters on the SRI. In this light, the specimen size should not be a major problem.

The effect of specimen size on the SRI was studied by, e.g. Kihlman and Nilsson [45]. The influence of the size was most significant in the sub-critical frequency range at individual edge and corner modes of the panel. The transmission due to sub-critical resonant vibration decreases with increasing area because the relative proportion of the perimeter decreases. For forced (non-resonant) transmission, the opposite behaviour was suggested by Sewell [46]. Above the critical frequency, the SRI does not depend on specimen size. However, the measurement results obtained by Michelsen [47] and Guy *et al.* [48] were not in agreement with previous theories. There does not seem to be an exhaustive answer to size effects. It seems to depend also on the wall structure. However, the influence of specimen size seems to be reasonably small, below 5 dB, being less significant than other factors, such as the niche effect discussed above.

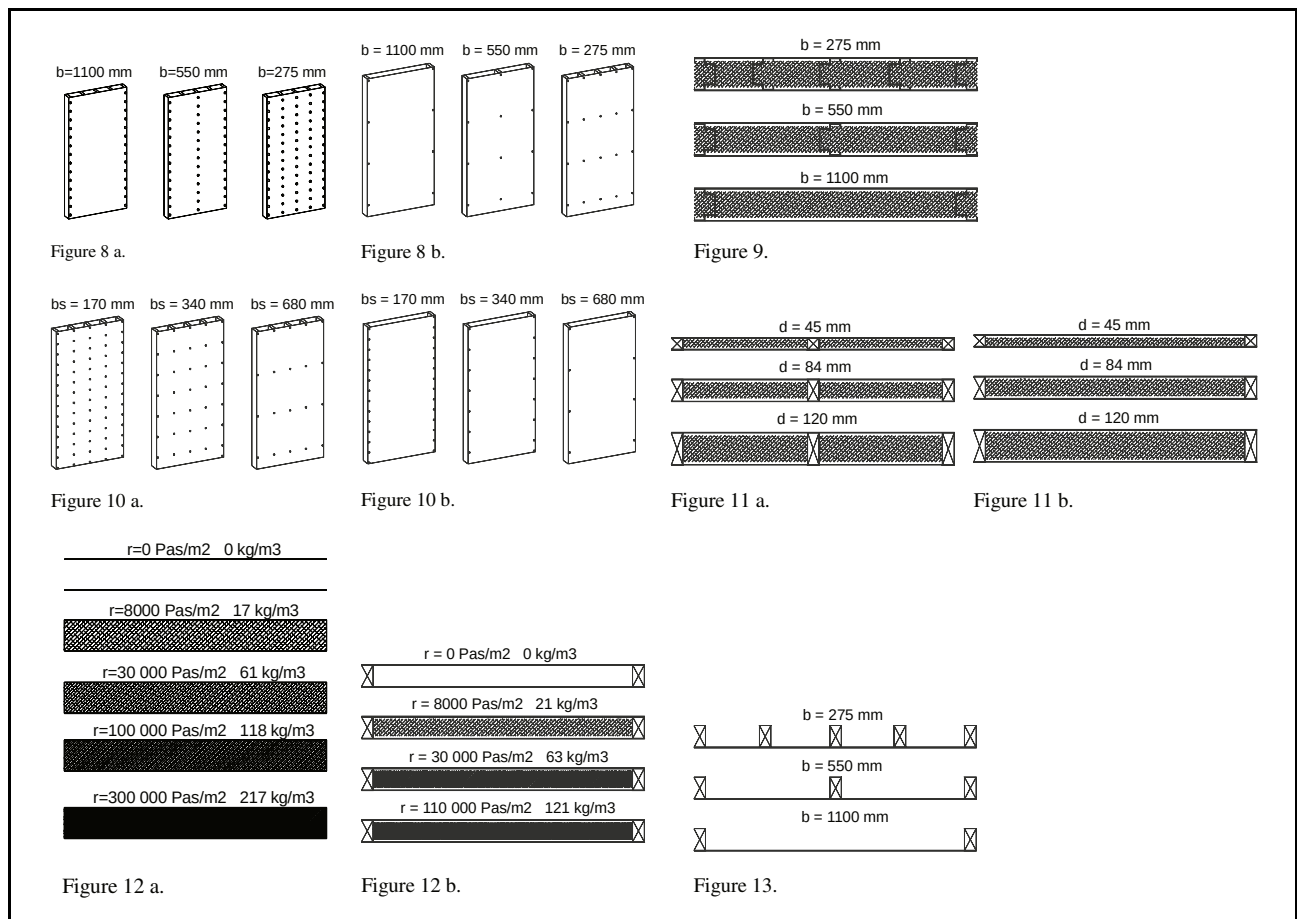


Figure 14. Continuation.

The effect of specimen size can be evident at low frequencies because the normal modes, predicted by equation (8), greatly depend on panel size. The frequency of the normal mode increases strongly with decreasing panel size. However, with steel plate of 2 mm thick, the value of  $f_{11}$  was 5 Hz with the smallest specimen dimension ( $L_x = 1100\text{ mm}$ ). Thus, the specimen size was not very critical in this respect. For heavy and stiff walls, the situation would be totally different. For example, for a 180 mm thick concrete wall (Young's modulus  $E = 2.6 \cdot 10^{10}\text{ Pa}$ ), the resonance frequency for a 1100 mm wide specimen is 450 Hz. In this case, the specimen size is very critical and the measurement results are applicable only for specimens of the same size and with the same boundary conditions.

It is well known that simple mass law does not predict the SRI at low frequencies. The SRI is typically larger at low frequencies than the field incidence mass law ( $20 \log m' f - 47\text{ dB}$ ) predicts. The effect is the stronger the smaller the specimen. According to Sewell [46] the SRI of forced and resonant vibration could be predicted reasonably well by normal incidence mass law ( $20 \log m' f - 42\text{ dB}$ ) reduced by  $10 \log[\ln(k\sqrt{S})]$ , where  $k$  (1/m) is the wave number and  $S$  is the specimen area (m<sup>2</sup>). Sewell's correction method is more accurate treatment of the radiation efficiency for forced waves of elements with finite dimensions. For example, if the specimen size is 2.5 m<sup>2</sup>, the

Sewell-corrected mass-law is approximately 9 dB, 6 dB and 3 dB higher than the field incidence mass law for an infinite specimen size at 50 Hz, 100 Hz and 1000 Hz, respectively. For a specimen size of 10 m<sup>2</sup>, the corrections are smaller, 4 dB, 2 dB and 1 dB at 50 Hz, 100 Hz and 1000 Hz, respectively. Recently, Kurra and Arditi confirmed this behaviour by simple experiments [49]. The theory seems to be valid because the slope decreases (bends) with decreasing frequency (see Figure 13). The theory of Sewell should hold also for double panels. However, the bending at low frequencies is difficult to see in Figures 4-12 because of other resonances. It can be concluded that the results would have been a couple of decibels higher at the lowest frequency bands if specimen size, 10 m<sup>2</sup> or larger, had been used.

There are also other low-frequency effects, like mode coupling between the rooms. In Figures 4-13, an increase of SRI at 63 and 80 Hz can occasionally be found. It is probably caused by the unfavourable coupling of room modes. The coupling was studied by measuring the SRI of an empty test opening, which should be approximately zero. The measured SRI of an empty test opening was  $0 \pm 2\text{ dB}$ , except for 10 dB at 63 Hz and 5 dB at 80 Hz.

The reproducibility of the measurement results is not completely self-evident. Recently, the results of an extensive round robin test were reported by Fausti *et al.* [50].

A total of 24 laboratories was involved in the round robin test, justifying significant statistical conclusions. The reproducibility was considerably poorer above 315 Hz than expected by ISO 140-2. The value of  $R_w$  varied between 47 and 52 dB for a double wall structure. For single panels the results were between 26 and 29 dB. Later, Smith *et al.* suggested that the lining material of the test aperture (built-in frame) was probably the principal factor causing the poor reproducibility values [51]. Laboratories with wooden and metal frames yielded, on average, smaller sound reduction indices than laboratories with concrete frames. Predictions with a SEA model agreed with this observation. The exclusion of laboratories with wooden frames from the statistical calculations decreased the reproducibility value considerably showing that the border coupling was one of the most important factors to be standardized. Problems are also caused by differences in laboratory structures. In particular, the mounting of the specimen and the design of the test aperture seem to have a strong influence on the result. More research is needed to obtain better agreement between present and new laboratories.

## 7. Conclusions

### 7.1. Uncoupled double panels (no studs)

The most important parameters for the SRI are the filling ratio of the absorbent and the cavity thickness. The SRI increases quite smoothly with increasing cavity thickness and the benefit is quite independent of frequency. The increase of SRI with increasing filling ratio is not linear and is independent of frequency. The increase of SRI is strongest when the cavity is empty and the first centimeters of absorbent are installed. The effect of the absorbent is negligible at lowest frequencies below the lowest cavity modes and largest at middle frequencies.

The effect of flow resistivity of the mineral wool in the range 8 000...300 000 Ns/m<sup>3</sup> played only a minor role.

Standing wave resonance of the cavity, perpendicular to the plate, could not be observed. This was not expected because the resonance is always mentioned in the literature. Instead, low order modes parallel to the plates, are very important. They occur typically at middle frequencies and can be effectively damped by adding absorbing material to the cavity.

### 7.2. Coupled double panels

The most important parameter is the stiffness of the stud. The sound insulation increases with decreasing dynamic stiffness in the middle and low frequencies. At high frequencies, the effect was smaller than expected. In overall, the frequency dependence of sound insulation with flexible studs was more complex than predicted by, e.g. Gu and Wang [13] or Davy [52]. It is probable that the previous models have been validated using too few experiments.

When the panels are attached to the rigid studs with screws, the screw spacing itself plays an important role.

Stud spacing plays an important role only at low frequencies. The studs divide the faceplate into narrower sub-panels, which have considerably high normal modes. These modes can cause even 20 dB dips in the SRI curve. With flexible studs, the stud spacing has only a small effect at low frequencies.

The thickness of the stud had a very small effect on SRI at middle and high frequencies. The effect was larger at low frequencies where the airborne path is dominating and stud path is negligible.

In the future, the origin of the low frequency resonances should be further investigated. It was suggested that they were caused by the normal modes of the sub-panels, which are formed between the studs when stud spacing was small. This model should be further investigated because the optimization of the low frequency region is of great importance whenever double walls are used. The effect of screw spacing and stud stiffness should be also more systematically investigated in the future because they are of major practical importance but the results obtained so far do not show very good consistency.

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## References

- [1] V. Hongisto: The calculation of the sound insulation of double panels – comparison of existing models. Proceedings of Internoise 2000, August 27-30, Nice, France, 2000, Vol 2, pp. 1243–1246.
- [2] V. Hongisto: Airborne sound insulation of wall structures - measurement and prediction methods. Doctoral Thesis, Helsinki University of Technology, Laboratory of Acoustics and Audio Signal Processing, Report 56, Finland, 2000.
- [3] A. London: Transmission of reverberant sound through double walls. *J. Acoust. Soc. Am.* **22** (1950) 270–279.
- [4] W. Loney: Effect of cavity absorption on the sound transmission loss of steel-stud gypsum wallboard partitions. *J. Acoust. Soc. Am.* **49** (1971) 385–390.
- [5] W. Loney: Effect of cavity absorption and multiple layers of wallboard on the sound-transmission loss of steel-stud partitions. *J. Acoust. Soc. Am.* **53** (1973) 1530–1534.
- [6] R. D. Ford, P. Lord, P. C. Williams: The influence of absorbent linings on the transmission loss of double-leaf partitions. *J. Sound Vib.* **5** (1967) 22–28.
- [7] P. P. Narang: Effect of fiberglass density and flow resistance on sound transmission loss of cavity plasterboard walls. *Noise Con. Eng. J.* **40** (1993) 215–220.
- [8] R. A. Novak: Sound insulation of lightweight double walls. *Appl. Acoust.* **37** (1992) 281–303.
- [9] A. C. K. Au, K. P. Byrne: On the insertion losses produced by acoustic lagging structures which incorporate flexurally

- orthotropic impervious barriers. *Acustica* **70** (1990) 284–291.
- [10] J. D. Quirt, A. C. C. Warnock: Influence of sound-absorbing material, stud type and spacing, and screw spacing on sound transmission through double-panel wall specimen. Proceedings of *Internoise 93*, Leuven, Belgium, August 24–26 1993, 971–974.
- [11] K. von Gösele, U. Gösele, B. Lakatos: Einfluß einer Gasfüllung auf die Schalldämmung von Isolierglasscheiben. *Acustica* **38** (1977) 167.
- [12] B. H. Sharp: A study of techniques to increase the sound insulation of building elements. Wyle laboratories report WR73-5, El Segundo, California, USA, 1973.
- [13] Q. Gu, J. Wang: Effect of resilient connection on sound transmission loss of metal stud double panel partitions. *Chinese J. Acoust.* **2** (1983) 113–126.
- [14] D. W. Green, C. W. Sherry: Sound transmission loss off gypsum wallboard partitions – Report #1. Unfilled steel stud partitions. *J. Acoust. Soc. Am.* **71** (1982) 90–96.
- [15] D. W. Green, C. W. Sherry: Sound transmission loss off gypsum wallboard partitions – Report #2. Steel stud partitions having cavities filled with glass fiber batts. *J. Acoust. Soc. Am.* **71** (1982) 902–907.
- [16] D. W. Green, C. W. Sherry: Sound transmission loss off gypsum wallboard partitions – Report #3. 2x4 in. wood stud partitions. *J. Acoust. Soc. Am.* **71** (1982) 908–914.
- [17] J. H. Rindel, D. Hoffmeyer: Influence of stud distance on sound insulation of gypsum board walls. Proceedings of *Internoise 91*, Sydney, Australia, 2–4 December 1991, 279–282.
- [18] R. J. M. Craik: Sound transmission through buildings using statistical energy analysis. Gower Publishing Ltd, Hampshire, England, 1996.
- [19] J. D. Quirt, A. C. C. Warnock, J. A. Birta: Summary report for consortium on gypsum board walls: sound transmission results. National Research Council Canada, Internal report IRC-IR-693, Institute for Research in Construction, Canada, October 1995.
- [20] R. J. M. Craik, R. S. Smith: Sound transmission through lightweight parallel plates, Part I: airborne sound. *Appl. Acoust.* **61** (2000) 223–245.
- [21] R. J. M. Craik, R. S. Smith: Sound transmission through lightweight parallel plates, Part II: structure-borne sound. *Appl. Acoust.* **61** (2000) 247–269.
- [22] L. Cremer, M. Heckl: Structure-borne sound (transl. E. E. Ungar) II. Edition, ch. 6. Springer-Verlag, Berlin, 1988.
- [23] R. J. M. Craik, R. S. Smith: Sound transmission through lightweight parallel plates, Part II: structure-borne sound. *Appl. Acoust.* **61** (2000) 247–269.
- [24] H. Muellner, I. Plotzlin: How “secondary” details of the construction and workmanship effect the airborne sound insulation of plasterboard walls. Proceedings of *Internoise*, The Hague, The Netherlands, 27–30 August, 2001, 1133–1136.
- [25] J. D. Quirt: Sound transmission through windows I. Single and double glazing. *J. Acoust. Soc. Am.* **72** (1982) 834–844.
- [26] J. D. Quirt: Sound transmission through windows II. Double and triple glazing. *J. Acoust. Soc. Am.* **74** (1983) 534–542.
- [27] V. Hongisto, J. Keränen, M. Lindgren: Sound insulation of doors - Part 1: Prediction models for structural and leak transmission. *Journal of Sound and Vibration* **230** (2000) 133–148.
- [28] V. Hongisto, J. Keränen, M. Lindgren: Sound insulation of doors - Part 2: Comparison between measurement results and predictions. *Journal of Sound and Vibration* **230** (2000) 149–170.
- [29] ISO 140-3:1995 (E) Acoustics: Measurement of sound insulation in buildings and of building elements – Part 3: Laboratory measurements of airborne sound insulation of building elements. International Organization for Standardization, Genève, Switzerland, 1995.
- [30] V. Hongisto, M. Lindgren, J. Keränen: Enhancing the maximum measurable sound reduction index of a laboratory using sound intensity technique and high amount of absorbers. *Journal of the Acoustical Society of America* **109** (2001) 254–265.
- [31] ISO 15186-1:2000 Acoustics: Measurement of sound insulation in buildings and of building elements using sound intensity, part 1: Laboratory conditions. International Organization for Standardization, Genève, Switzerland, 2000.
- [32] ISO 140-2:1991 (E) Acoustics: Measurement of sound insulation in buildings and of building elements – Part 2: Determination, verification and application of precision data. International Organization for Standardization, Genève, Switzerland, 1991.
- [33] F. Jacobsen, J. H. Rindel: Time reversed decay measurements. *J. Sound Vib.* **115** (1987) 163–170.
- [34] D. Rife, J. Vanderkooy: Transfer-function measurement with maximum-length sequences. *J. Audio Eng. Soc.* **37** (1989) 419–444.
- [35] A. Meier, A. Schmitz: Application of total loss factor measurements for the determination of sound insulation. *Building Acoustics* **6** (1999) 71–84.
- [36] ASTM C 522-87: Standard test method for airflow resistance of acoustical materials. American Society for Testing and Materials, Reapproved 1997.
- [37] ISO 9053-1:1991 (E) Acoustics: Materials for acoustical applications - Determination of airflow resistance. International Organization for Standardization, Genève, Switzerland, 1989.
- [38] ISO 9052-1:1989 (E) Acoustics: Determination of dynamic stiffness – Part 1: Materials used under floating floors in dwellings. International Organization for Standardization, Genève, Switzerland, 1989.
- [39] C. Hopkins, R. Wilson, R. J. M. Craik: Dynamic stiffness as an acoustic specification parameter for wall ties used in masonry cavity walls. *Appl. Acoust.* **58** (1999) 51–68.
- [40] A. Cops, M. Minten, H. Myncke: Influence of the design of transmission rooms on the sound transmission loss of glass - intensity versus conventional method. *Noise Con. Eng. J.* **28** (1987) 121–129.
- [41] R. W. Guy, P. Sauer: The influence of sills and reveals on sound transmission loss. *Appl. Acoust.* **17** (1984) 453–476.
- [42] A. Cops, M. Minten: Comparative study between the sound intensity method and the conventional two-room method to calculate the sound transmission loss of wall constructions. *Noise Con. Eng. J.* **22** (1984) 104–111.

- [43] F. J. Fahy: Sound and structural vibration – Radiation, transmission and response, ch. 4. Academic Press Ltd., London, England, 1985.
- [44] K. A. Mulholland, H. D. Parbrook, A. Cummings: The transmission loss of double panels. *J. Sound Vib.* **6** (1967) 324–334.
- [45] T. Kihlman, A. C. Nilsson: The effects of some laboratory designs and mounting conditions on reduction index measurements. *J. Sound Vib.* **24** (1972) 349–364.
- [46] E. C. Sewell: Transmission of reverberant sound through a single-leaf partition surrounded by an infinite rigid baffle. *J. Sound Vib.* **12** (1970) 21–32.
- [47] N. Michelsen: Effect of size on measurements of the sound reduction index of a window or a pane. *Appl. Acoust.* **16** (1983) 215–234.
- [48] R. W. Guy, A. De Mey, P. Sauer: The effect of some physical parameters upon the laboratory measurements of sound transmission loss. *Appl. Acoust.* **18** (1985) 81–98.
- [49] S. Kurra, D. Arditi: Determination of sound transmission loss of multilayered elements Part 2: an experimental study. *Acta Acustica–Acustica* **87** (2001) 592–603.
- [50] P. Fausti, R. Pompoli, R. Smith: An intercomparison of laboratory measurements of airborne sound insulation of lightweight plasterboard walls. *J. Build. Acoust.* **6** (1999) 127–140.
- [51] R. S. Smith, R. Pompoli, P. Fausti: An investigation into the reproducibility values of the European inter-laboratory test for lightweight walls. *J. Build. Acoust.* **6** (1999) 187–210.
- [52] J. L. Davy: Predicting the sound insulation of stud walls. *Proceedings of Inter-Noise 1991, Sydney, Australia, 2-4 December 1991, Volume 1*, pp. 251–254.